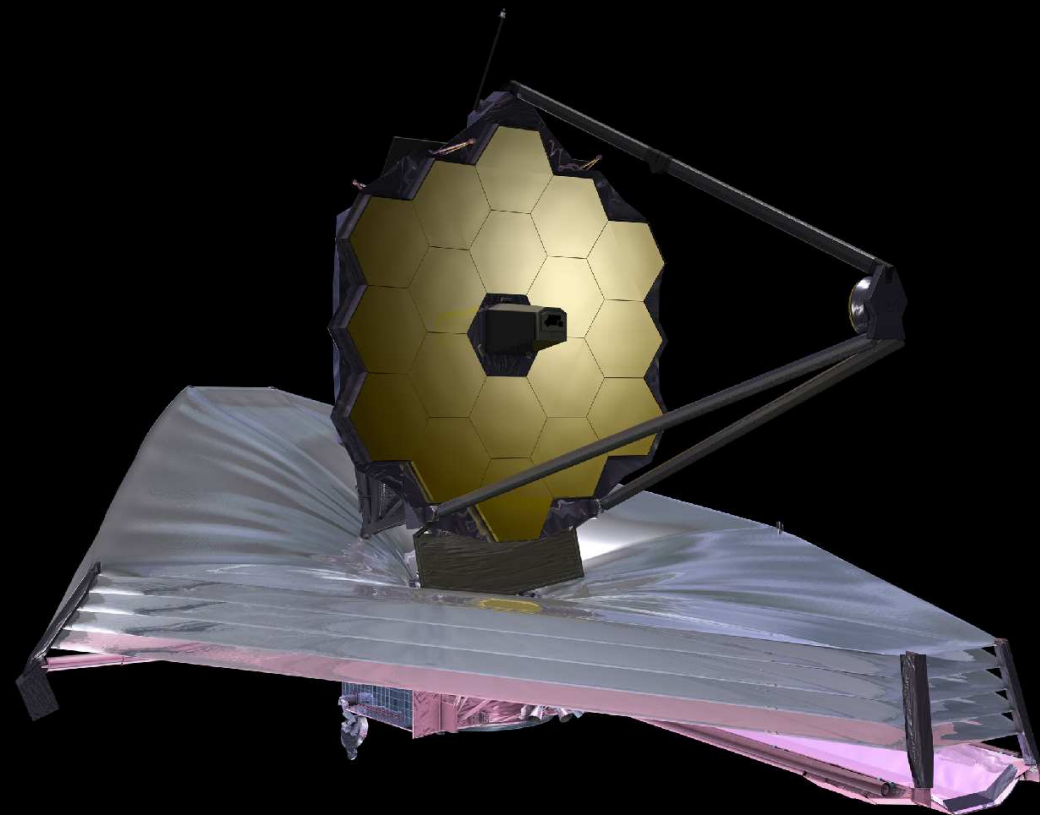


Using Hubble Object Finding and Classification Software to Accurately Measure Cancer Cell Spreading and Type 2 Diabetes Markers in an Early Stage

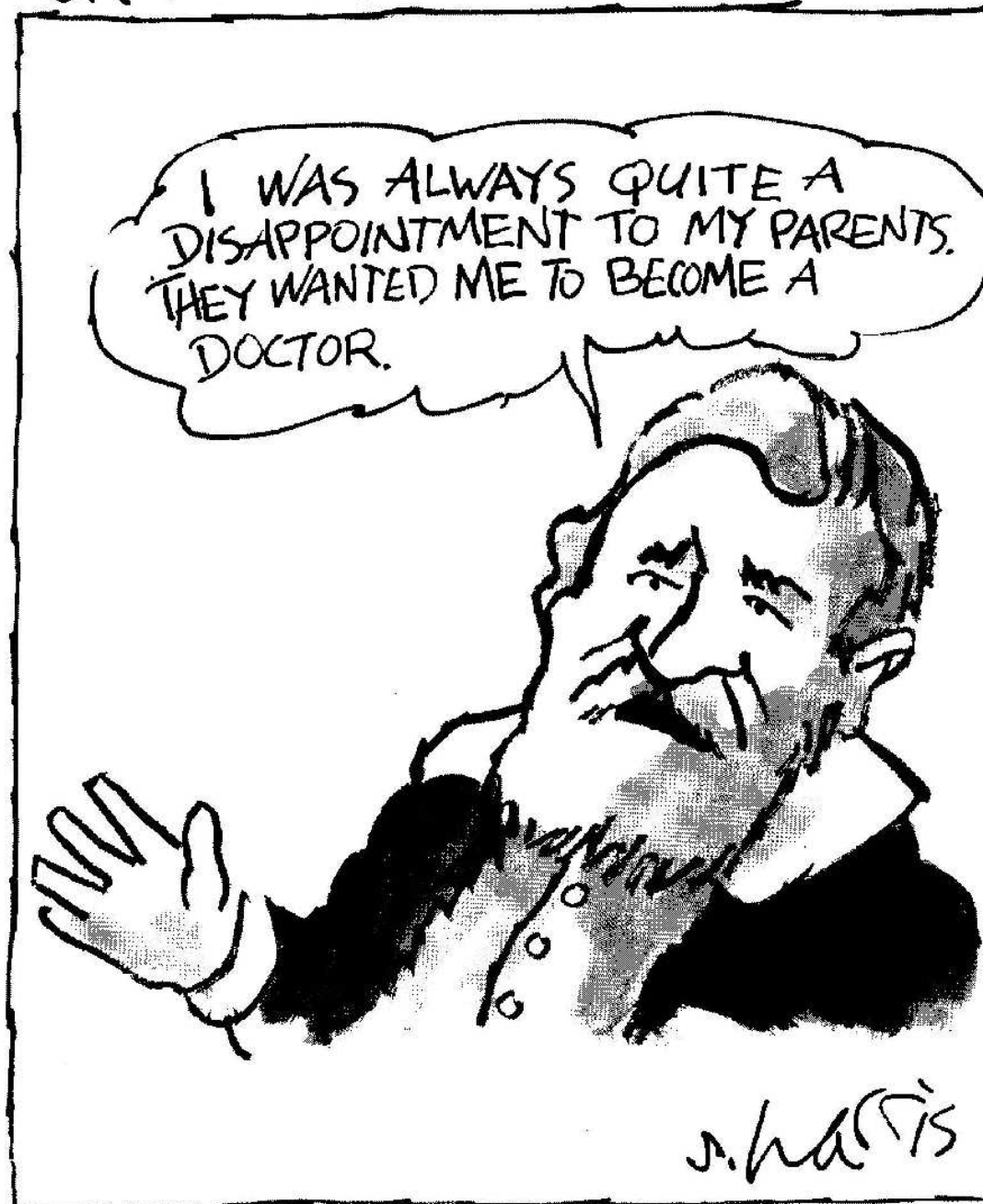
Rogier Windhorst (School of Earth and Space Exploration, ASU)

P. Baluch, L. Burnett, J. Brower, A. Casano, J. Olson (ASU), R. Herman, & J. Targovnik (Banner Health), R. Ryan, K. Tamura, V. Taylor (ASU)



Research Summary at Santé Ventures Meeting, ASU Skysong, Thursday Dec. 8, 2011, Tempe, AZ

GALILEO'S BURDEN



Why I personally got involved in medical research ... (true story!)

Outline

(1) Summary of the NASA Hubble Space Telescope connection:

- (1a) NASA missions to study Cosmic Dawn, First Light, Dark Ages.
- (1b) NASA Hubble Space Telescope (HST) image analysis.

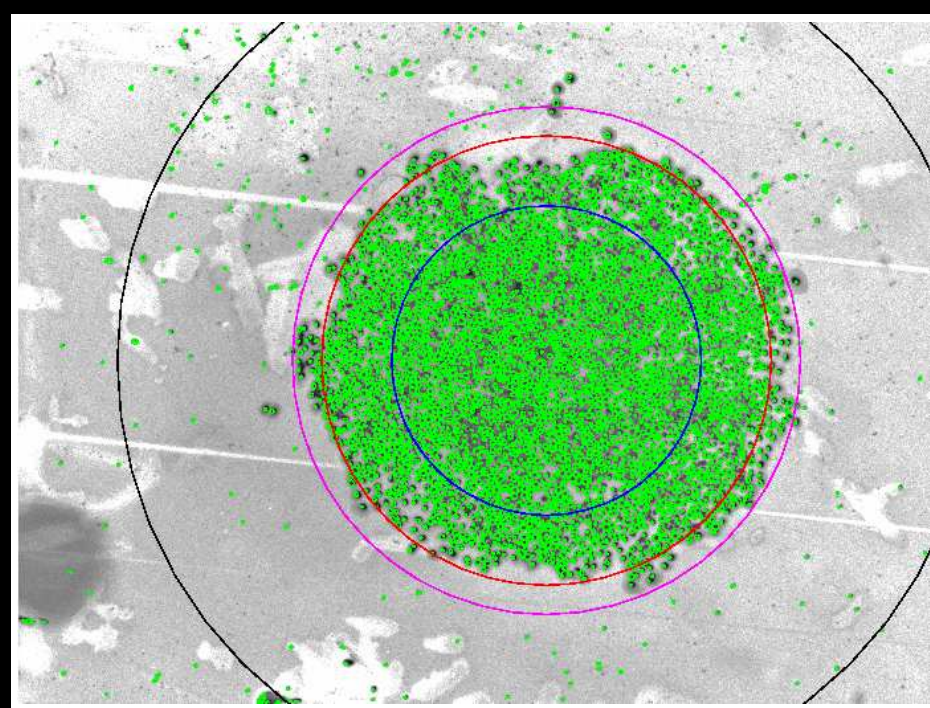
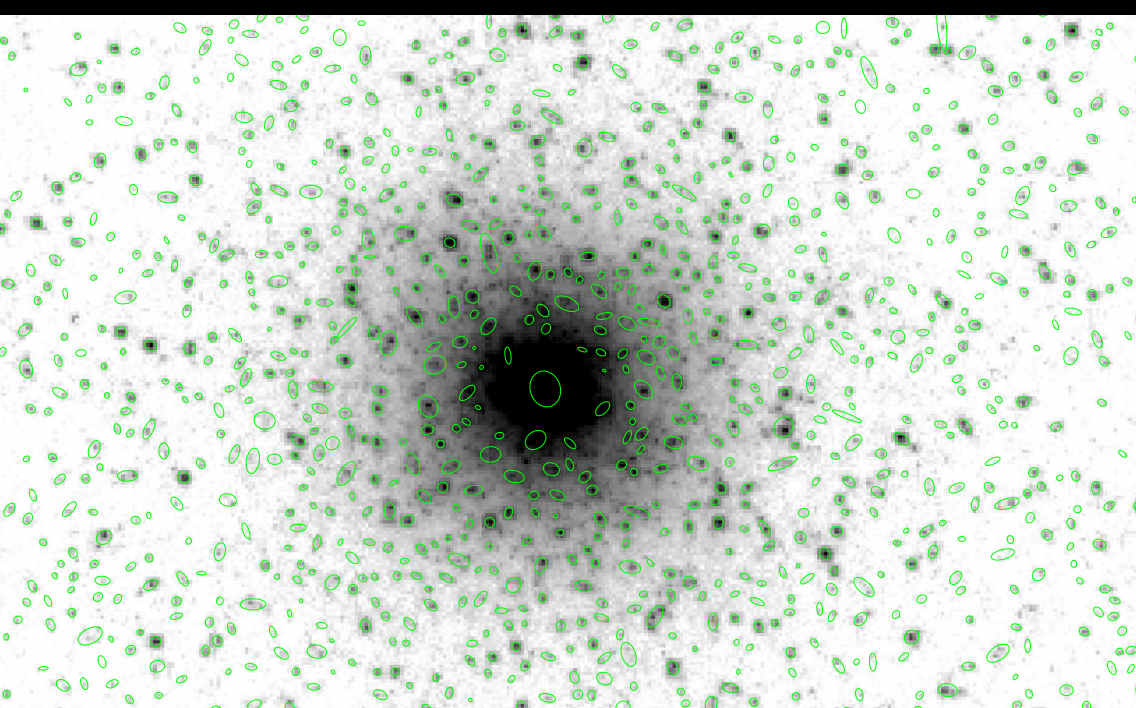
(2) Using Hubble Object Finding and Classification Software to Find Cancer Cell Spreading and Markers for Diabetes Type 2 in an Early Stage:

- (2a) The Object Finding, Detection, Deblending, and Classification method.
- (2b) Application of Hubble Software to Study Spreading of Cancer Cells.
- (2c) Application to C-fiber Skin-biopsies to Normal and Obese Patients.

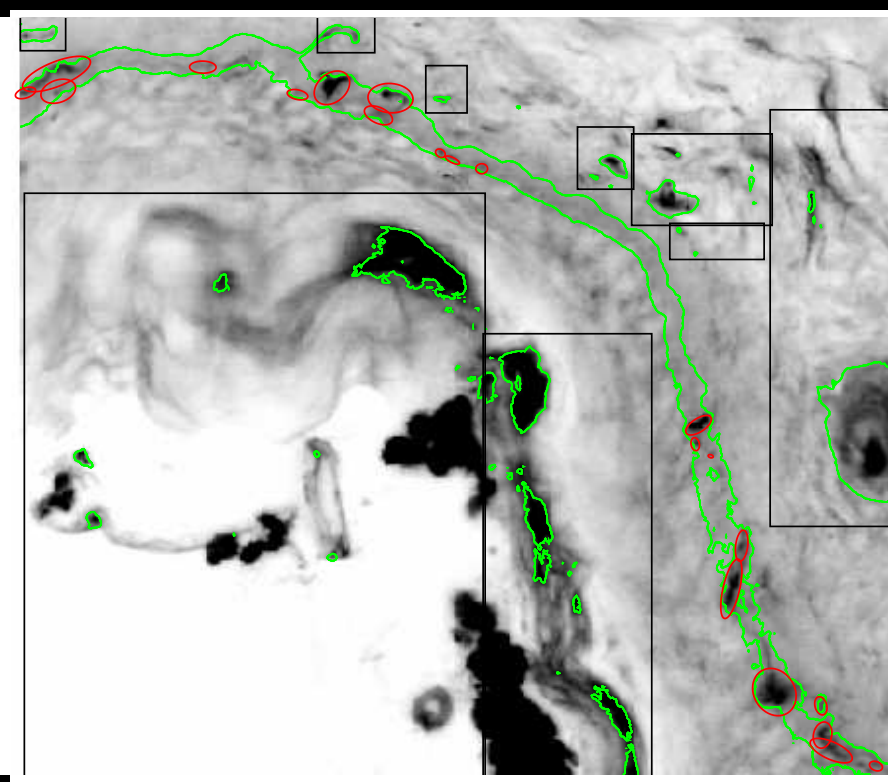
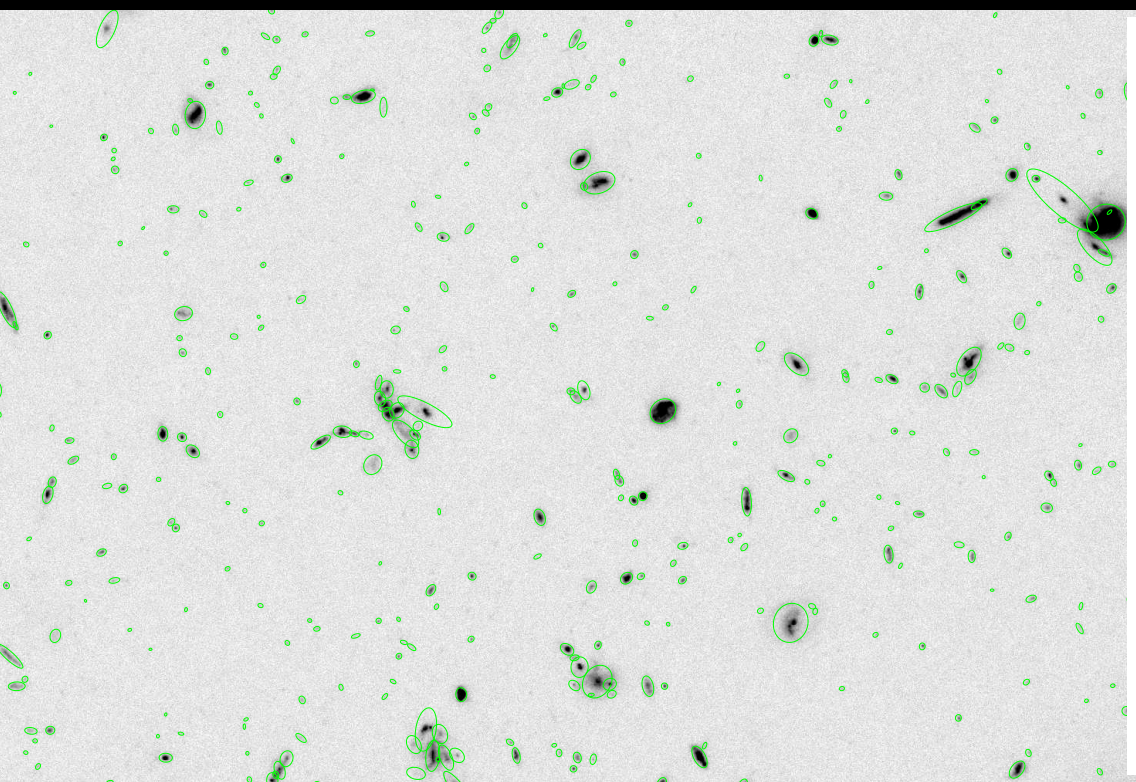
(3) Results and and Conclusions:

- Reliably find & measure faint features in non-uniform, noisy background.
- Reliably process large numbers of images at once (msec of CPU time).

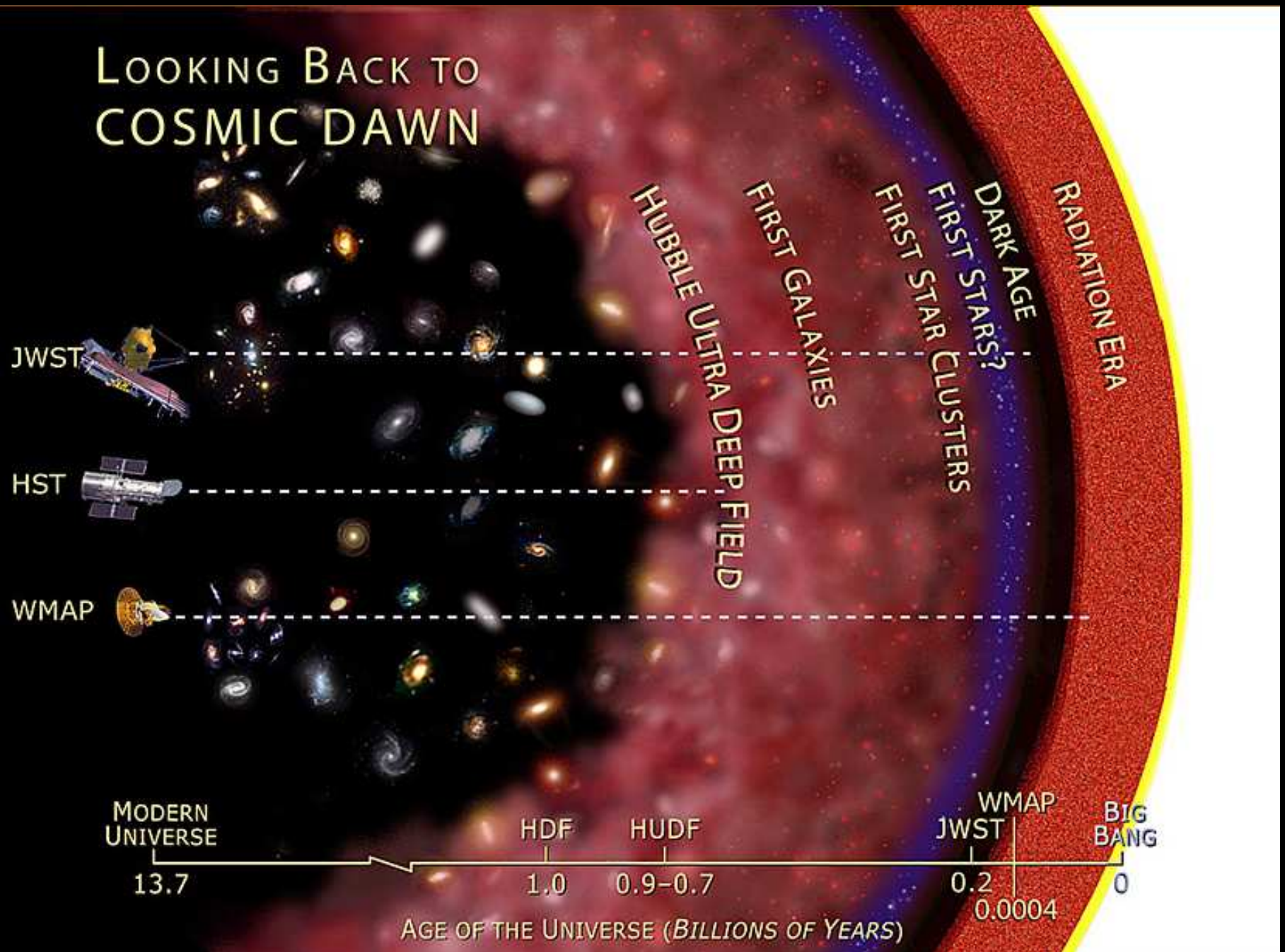
Sponsored by NASA/ASU/Banner Health



Hubble star-cluster & galaxies (left); Bladder cancer & C-fiber biopsies (right).



(1a) NASA missions that study Cosmic Dawn, First Light, & the Dark Ages



Looking back in time $\Rightarrow$ Deal with many overlapping galaxy images!

Intro: Cosmic Expansion and Contents of the Universe

Expansion $\Rightarrow$ redshift

$$\lambda_{obs} = \lambda_{rest} \cdot (1+z)$$

Hubble's Law:

$$D \simeq v / H_0 \simeq (c/H_0) \cdot z = R_0 \cdot z$$

Cosmic Content:

inside $R_0 = (c/H_0) \simeq 13.73$ Glyr:

$$[t_{univ} = (211 \pm 2) \times (t_{dino} = 65 \text{ Myr})]$$

Photons (light):

$$N_{h\nu} \sim 10^{89}$$

Baryons (atoms):

$$N_b \sim 10^{80}$$

η = Photons/Baryons

$$\eta \sim 10^9 \implies \text{He/H ratio} = 0.235$$

Energy Density:

as fraction of critical closure density:

Baryons (atoms):

$$\Omega_b = \rho_b / \rho_{crit} \simeq 0.042$$

Dark Matter:

$$\Omega_d = \rho_d / \rho_{crit} \simeq 0.20$$

Dark Energy (Λ):

$$\Omega_\Lambda = \rho_\Lambda / \rho_{crit} \simeq 0.76$$

(Supermassive) black holes:

$$\rho_{smbh} / \rho_{crit} \simeq 0.0001$$

Total

$$\Omega_{tot} = \rho_{tot} / \rho_{crit} \simeq 1.00 \pm 0.02$$

Corollary: 40 years S/W experience to find, deblend, & classify faint objects.

(1) The NASA Hubble Space Telescope connection

- [illegible]

[illegible]

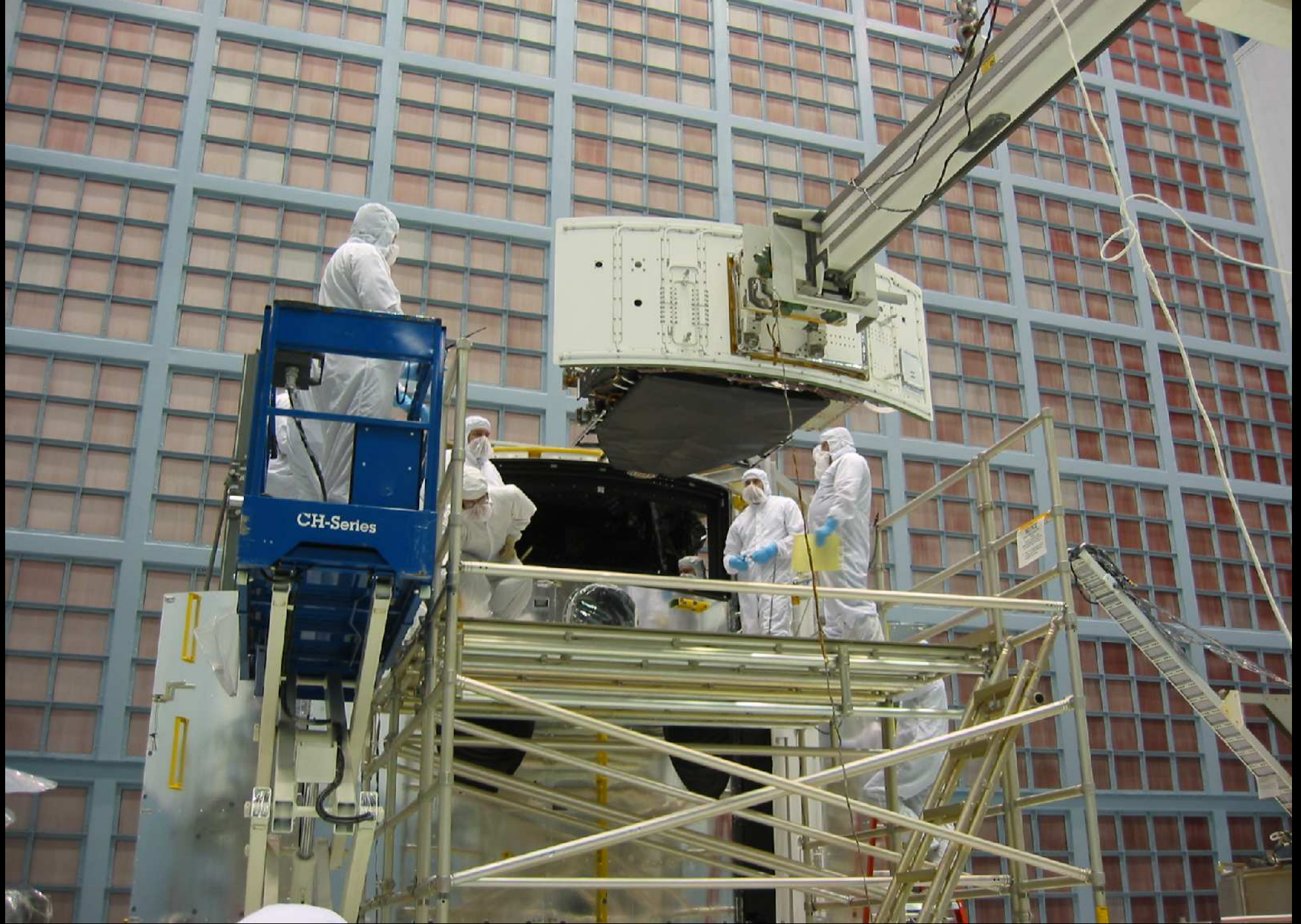
This project of medical applications of Hubble galaxy classification software builds on a 40 year history of astronomical software development.

(1) Recent key aspects of the Hubble Space Telescope (HST) project:



The HST Advanced Camera for Surveys (ACS) — launched 2002 (SM3B).



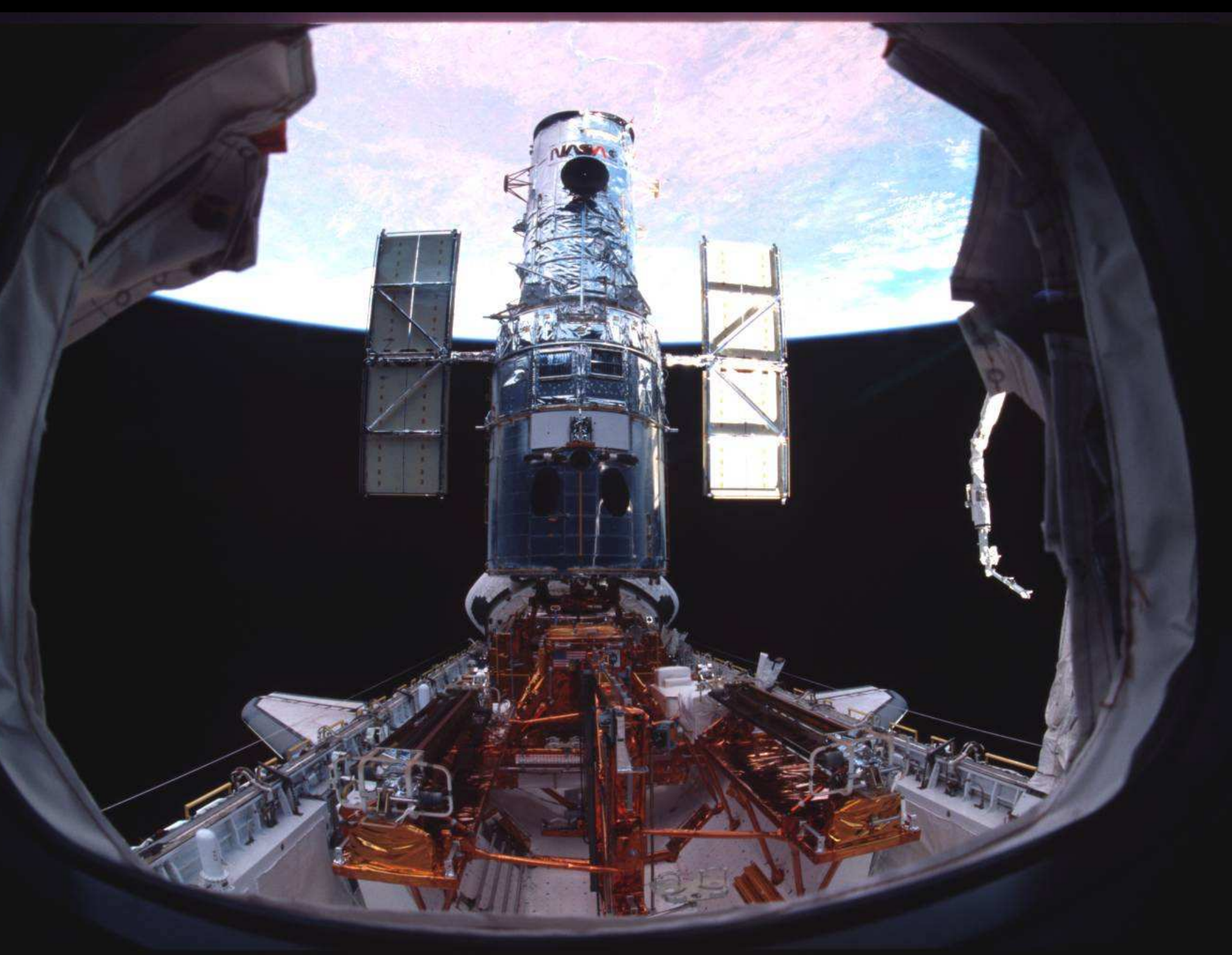


Wide Field Camera 3 for SM4 in 2009: More powerful HST imaging than ever.

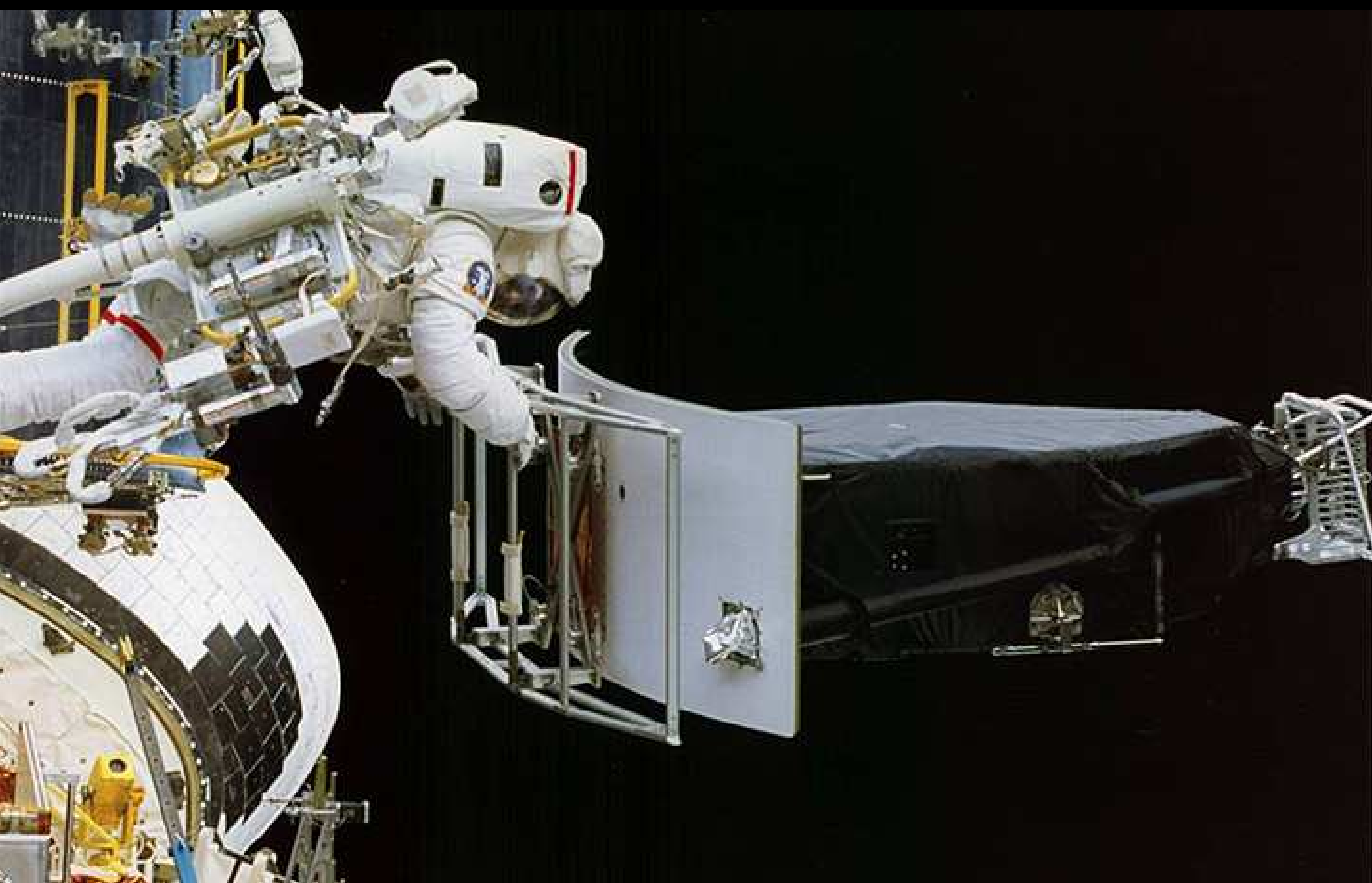












Installing Wide Field Camera 2 (WFPC2) during SM1 in December 1993.
Similar to what astronauts did with WFC3 during SM4 in May 2009.



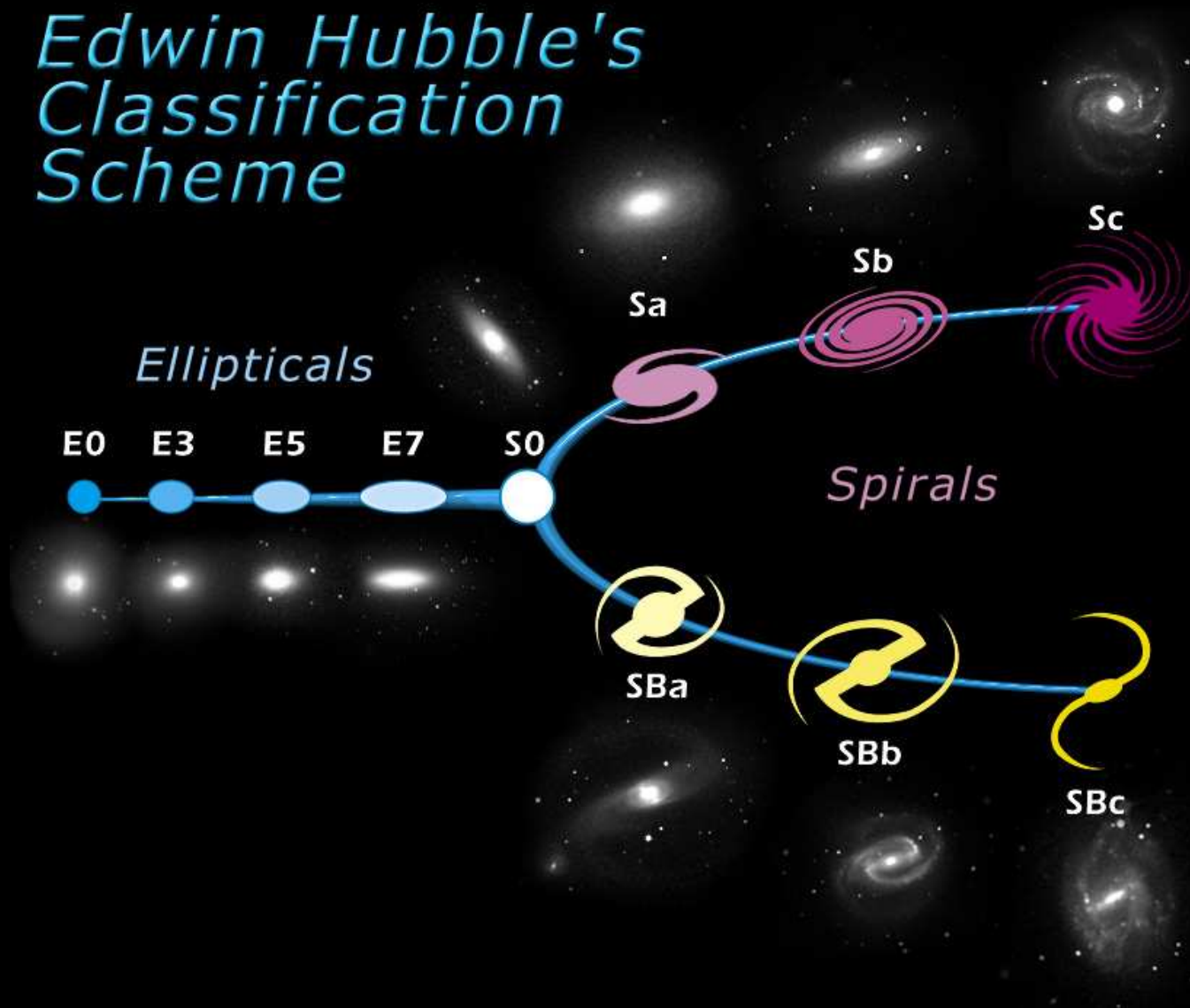
(2) New studies of the Cosmos with the Hubble Wide Field Camera 3



Wide Field Camera 3 for SM4 in 2009: More powerful HST imaging than ever.

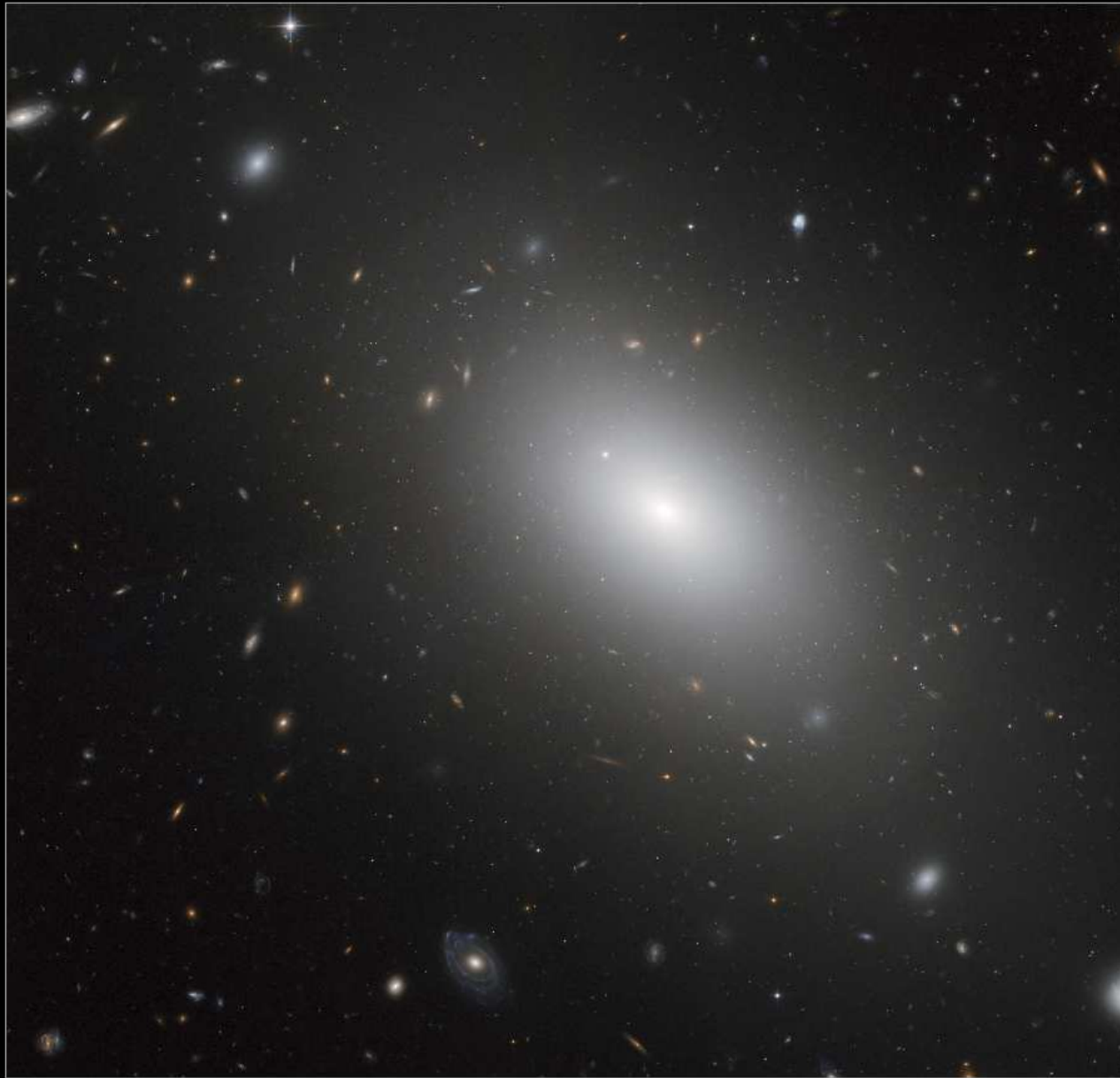
(2) HST turned the classical Hubble sequence upside down!

Edwin Hubble's Classification Scheme



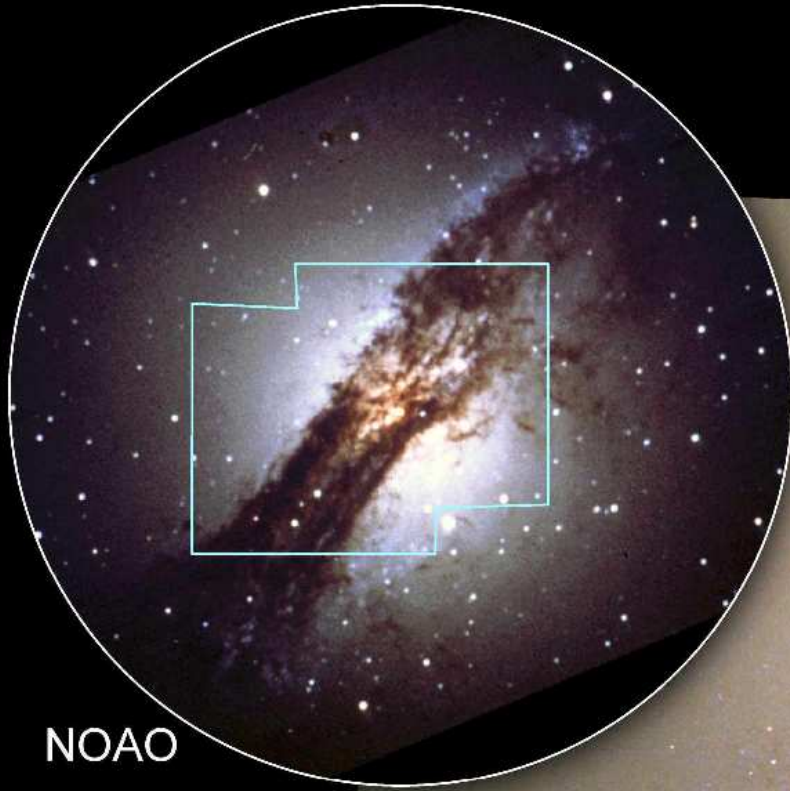
Who (when)	Cosmic Epoch	Ellipticals	Spirals	Irr's/mergers
Hubble (1920's)	$z=0$ (13.73 Gyr)	$\sim 40\%$	$\gtrsim 50\%$	$\lesssim 10\%$
HST (1990's)	$z \simeq 1-2$ (3-6 Gyr)	$\lesssim 15\%$	$\sim 30\%$	$\gtrsim 55\%$!

Elliptical Galaxy NGC 1132



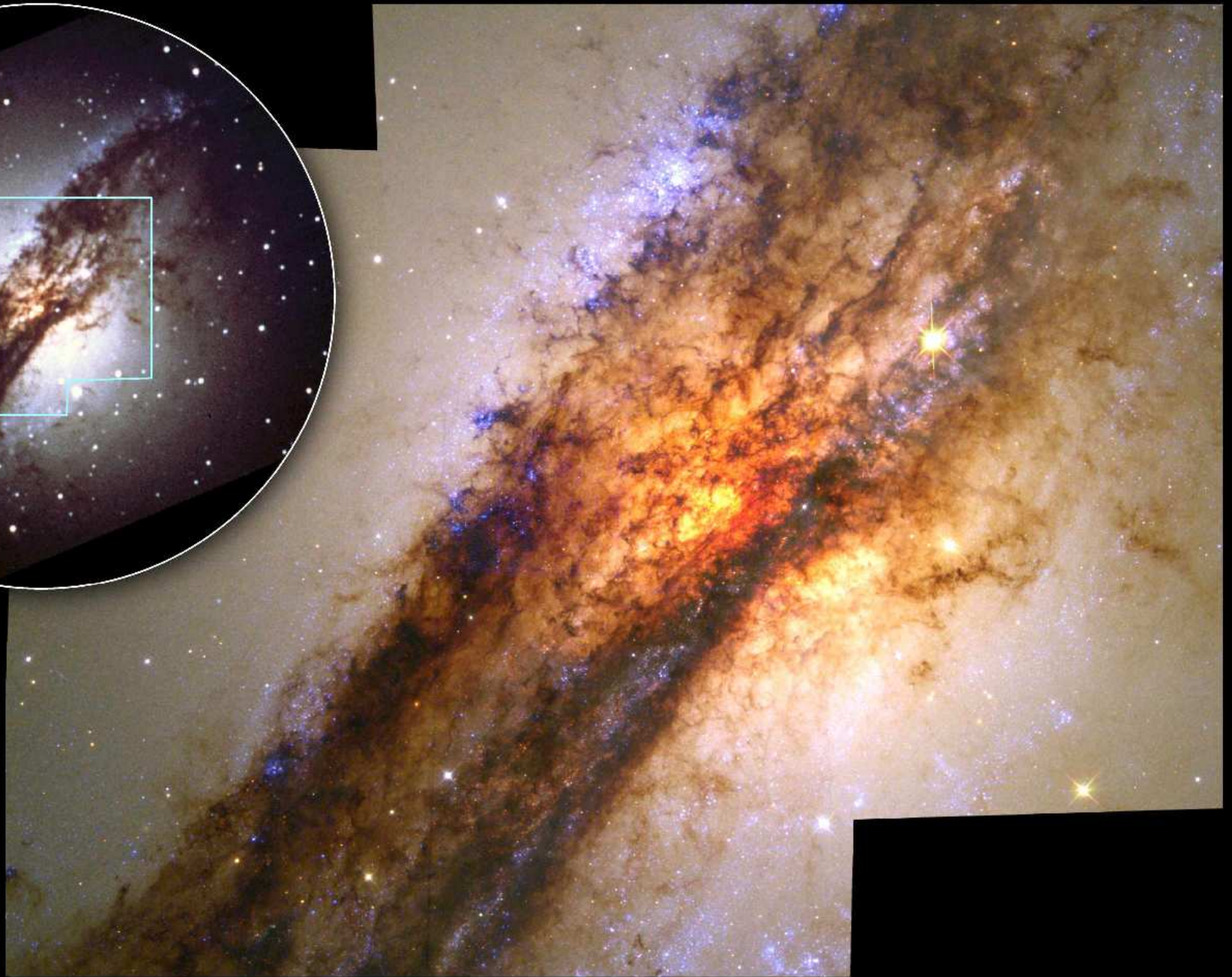
Hubble
Heritage

M. Rutkowski (Dissertation).



NOAO

HST



Active Galaxy Centaurus A

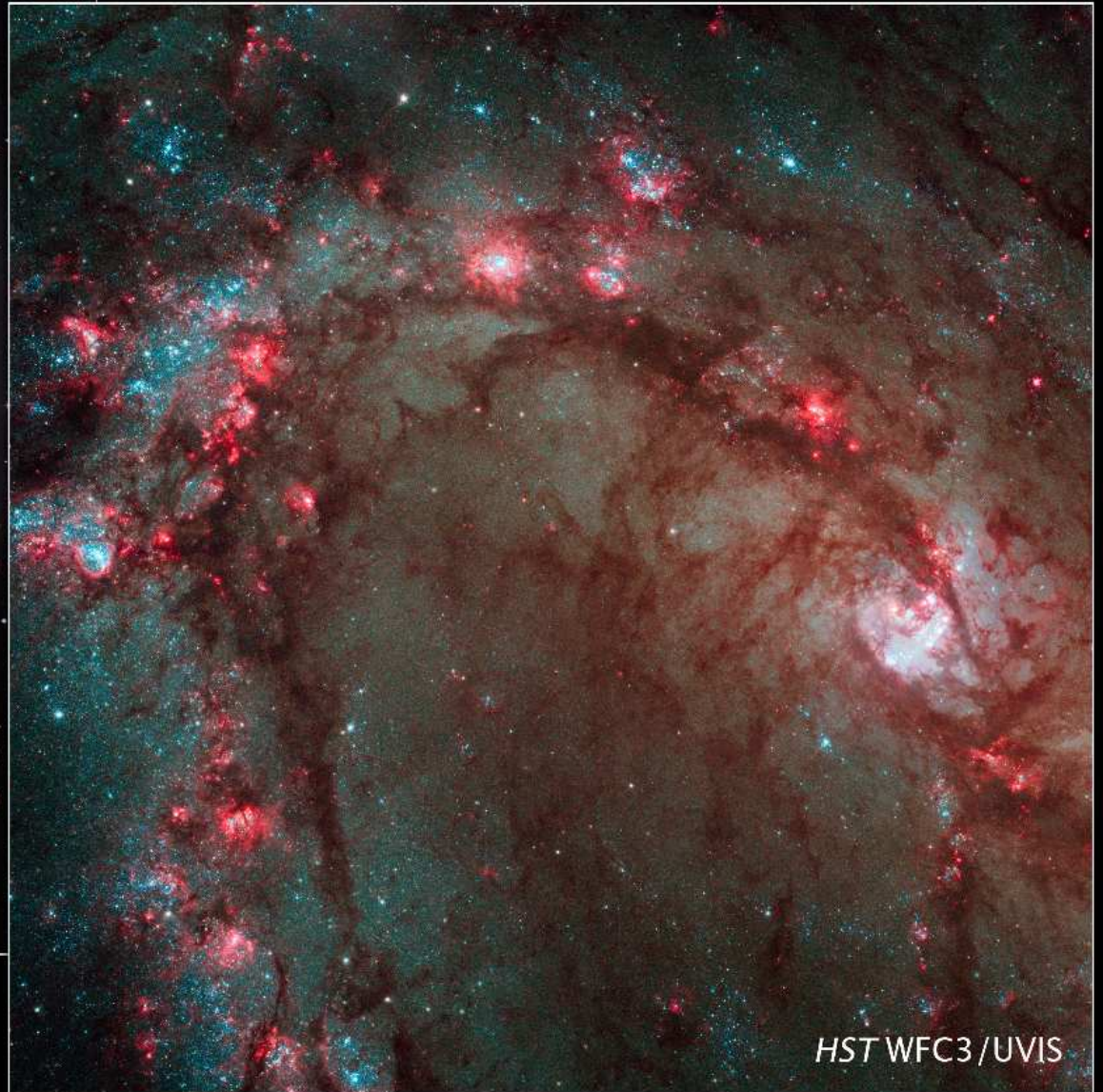
Hubble Space Telescope • Wide Field Planetary Camera 2

Centaurus A
NGC 5128
HST WFC3/UVIS

F225W+F336W+F438W
F487N H β
F502N [O III]
F547M γ
F657N H α + [N II]
F673N [S II]
F814W I

3000 light-years
1400 parsecs 56''



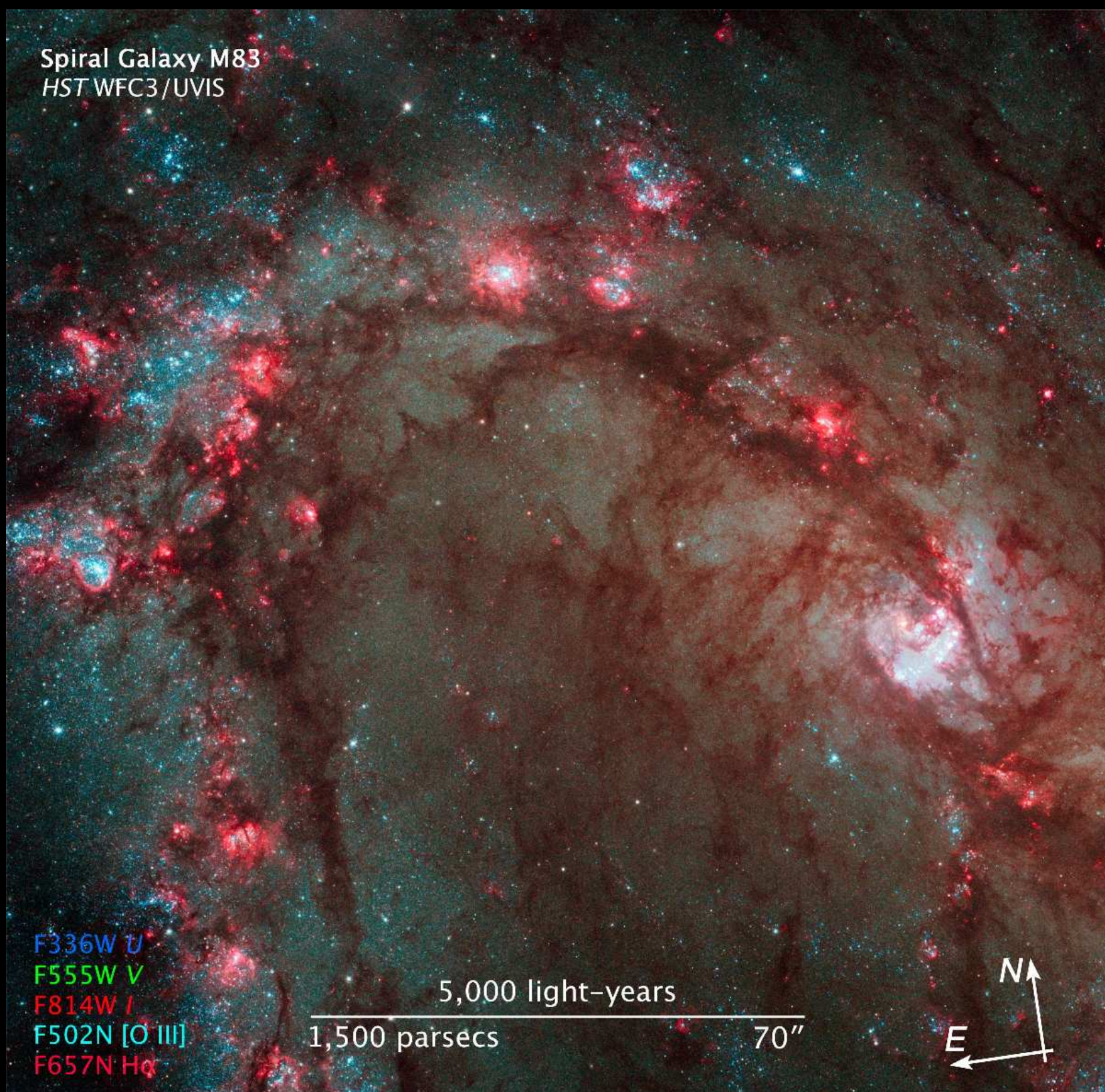


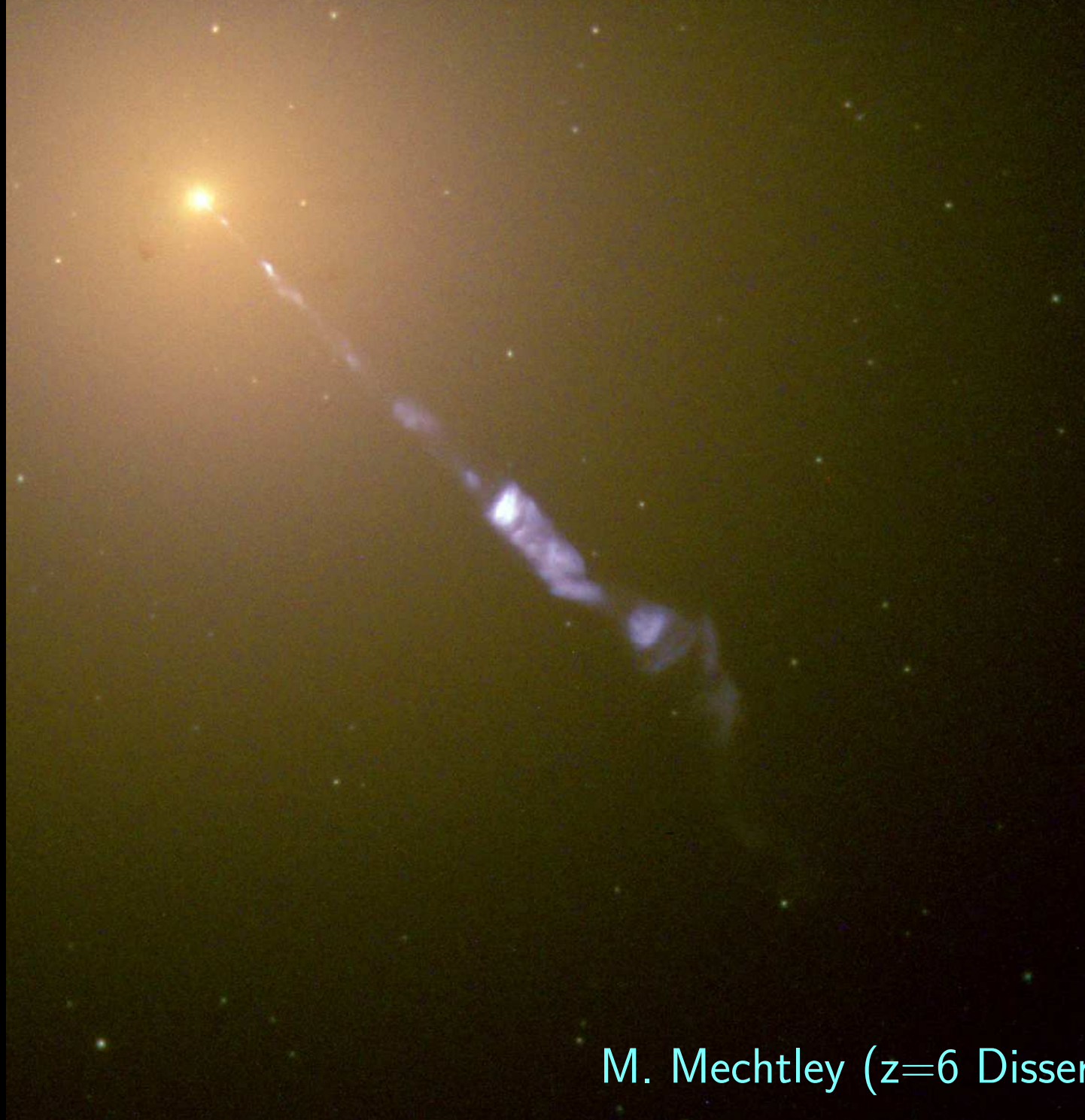
Spiral Galaxy M83
Hubble Space Telescope ■ WFC3/UVIS

Spiral Galaxy M83
HST WFC3/UVIS

F336W U
F555W V
F814W I
F502N [O III]
F657N H α

5,000 light-years
1,500 parsecs 70"





M. Mechtley (z=6 Dissertation).

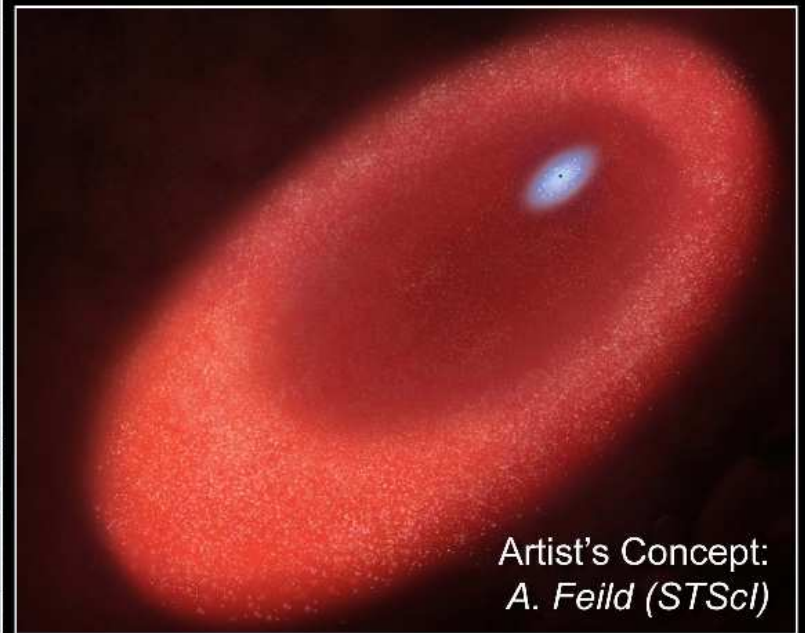
Elliptical galaxy M87 with Active Galactic Nucleus (AGN) and relativistic jet.



Photo Copyright R. Gendler



*HST WFPC2 image:
T. Lauer (NOAO/AURA/NSF)*



*Artist's Concept:
A. Feild (STScI)*

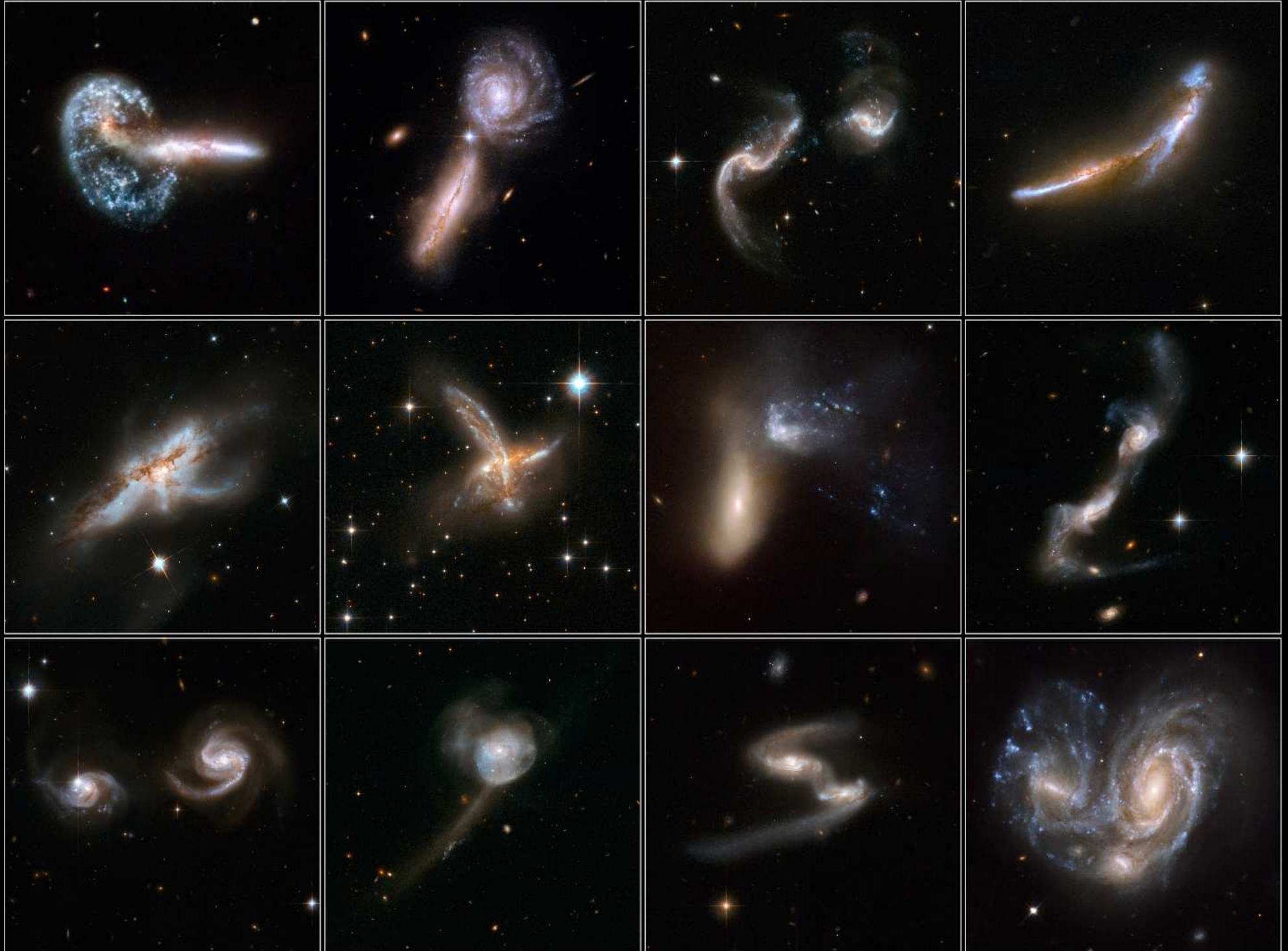
Andromeda Galaxy Nucleus ■ M31
Hubble Space Telescope ■ WFPC2

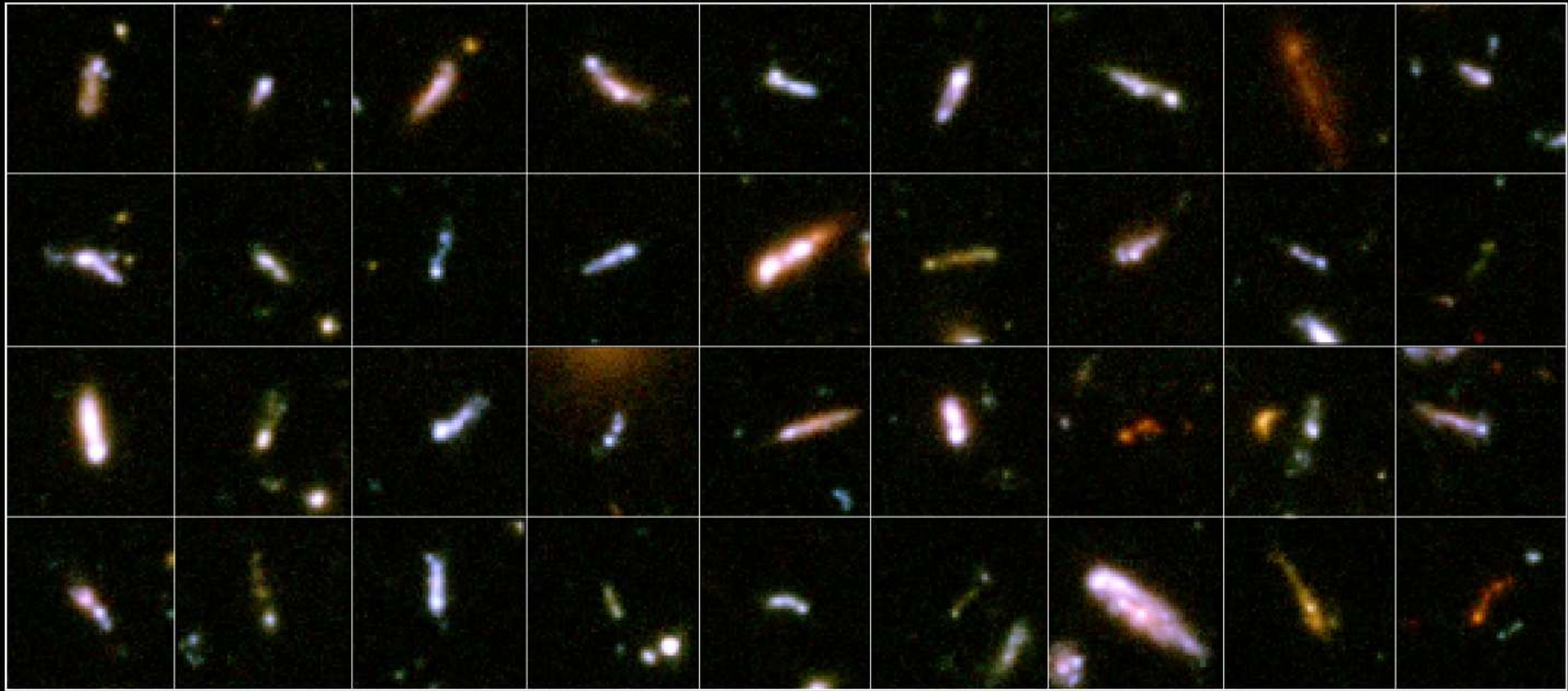


HST Antenna galaxy: Prototype of high redshift, star-forming, major merger?

Interacting Galaxies

Hubble Space Telescope • ACS/WFC • WFPC2





“Tadpole” Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope ■ ACS/WFC

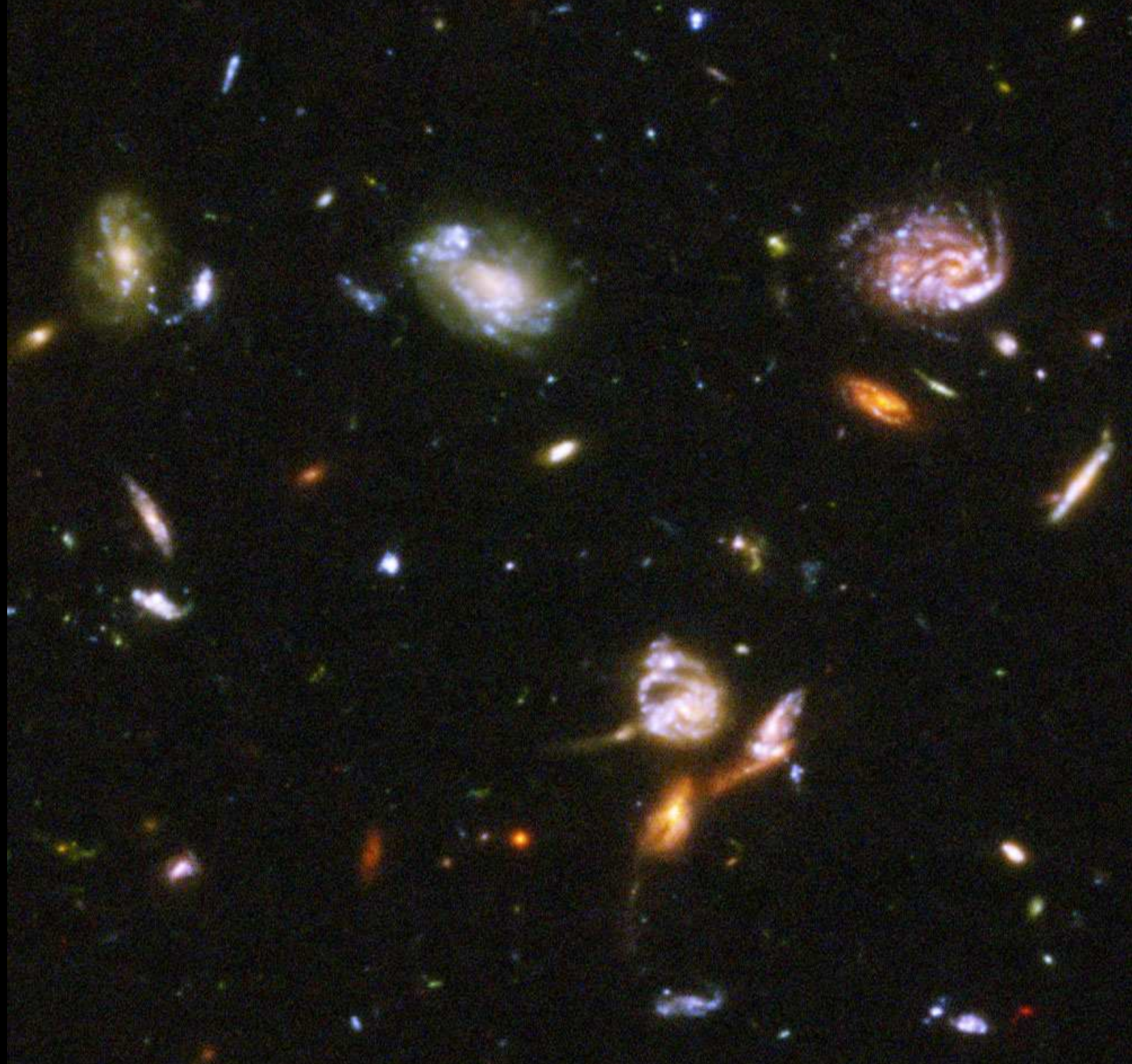
NASA, ESA, A. Straughn, S. Cohen and R. Windhorst (Arizona State University), and the HUDF team (STScI)

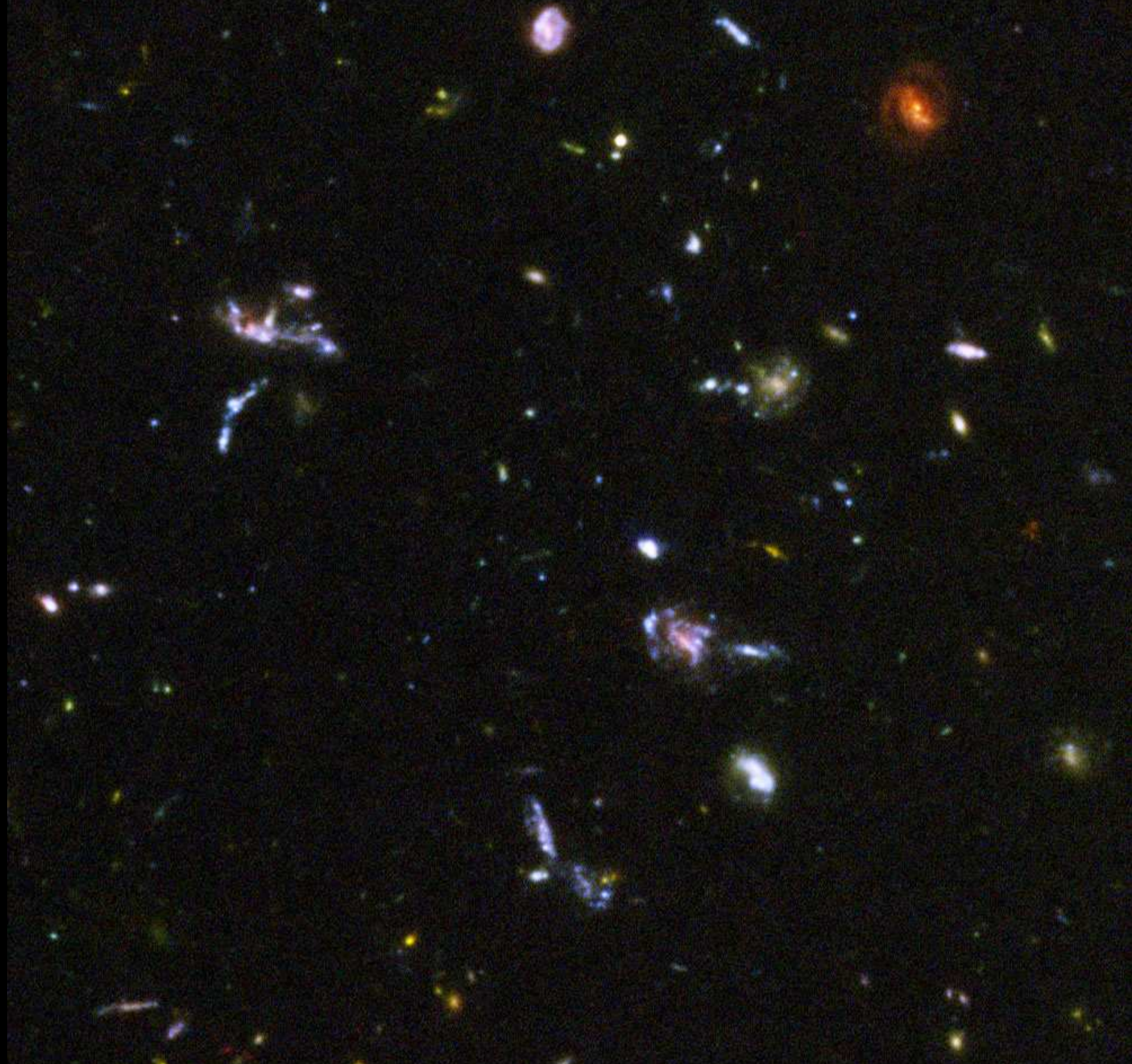
STScI-PRC06-04

Merging galaxies constitute $\lesssim 1\%$ of Hubble sequence today (age $\gtrsim 12.5$ Gyr).

Tadpole galaxies are early stage mergers, very common at $z \gtrsim 2$ (age $\lesssim 3$ Gyr).

JWST will measure galaxy assembly to $z \lesssim 20$ (age $\simeq 0.2$ Gyr).

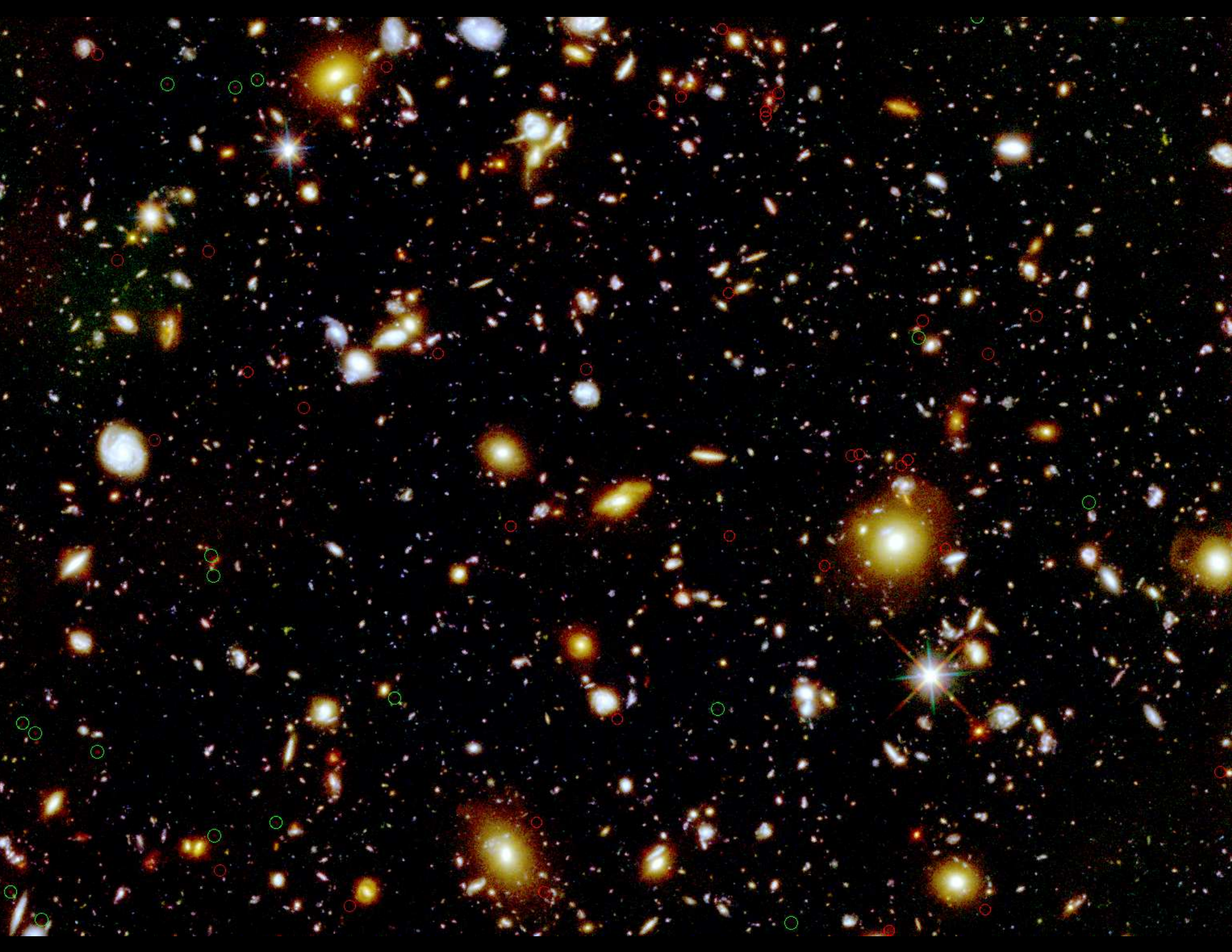




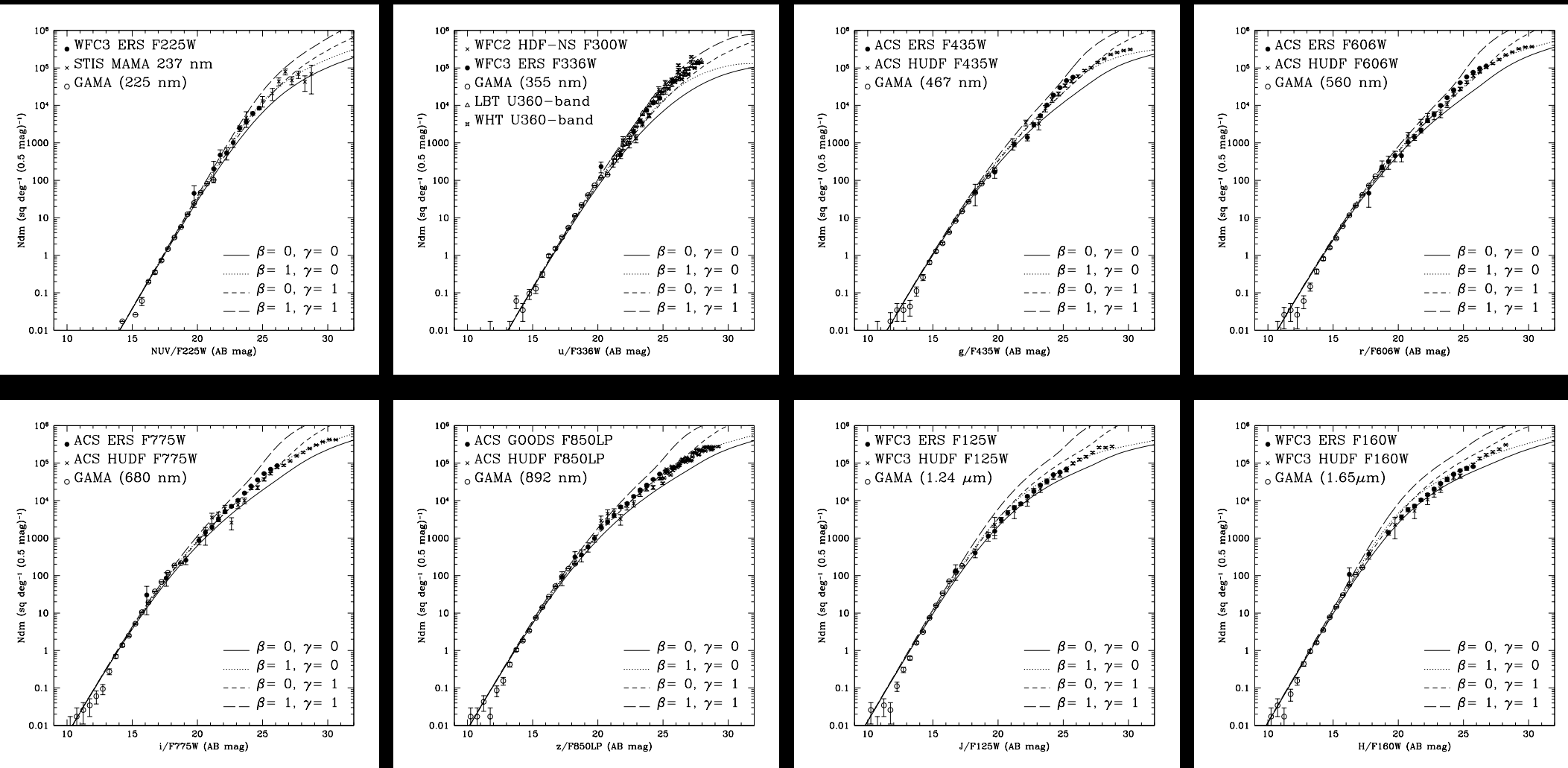


HST/WFC3 & ACS reach $AB=26.5-27.0$ mag (100 fireflies from Moon) over $0.1 \times$ full Moon area in 10 filters from $0.2-2\mu\text{m}$ wavelength, with $0.07-0.15''$ FWHM resolution (=dime on US Capitol seen from Baltimore).

Images so crowded that properly finding/deblending faint objects is essential.



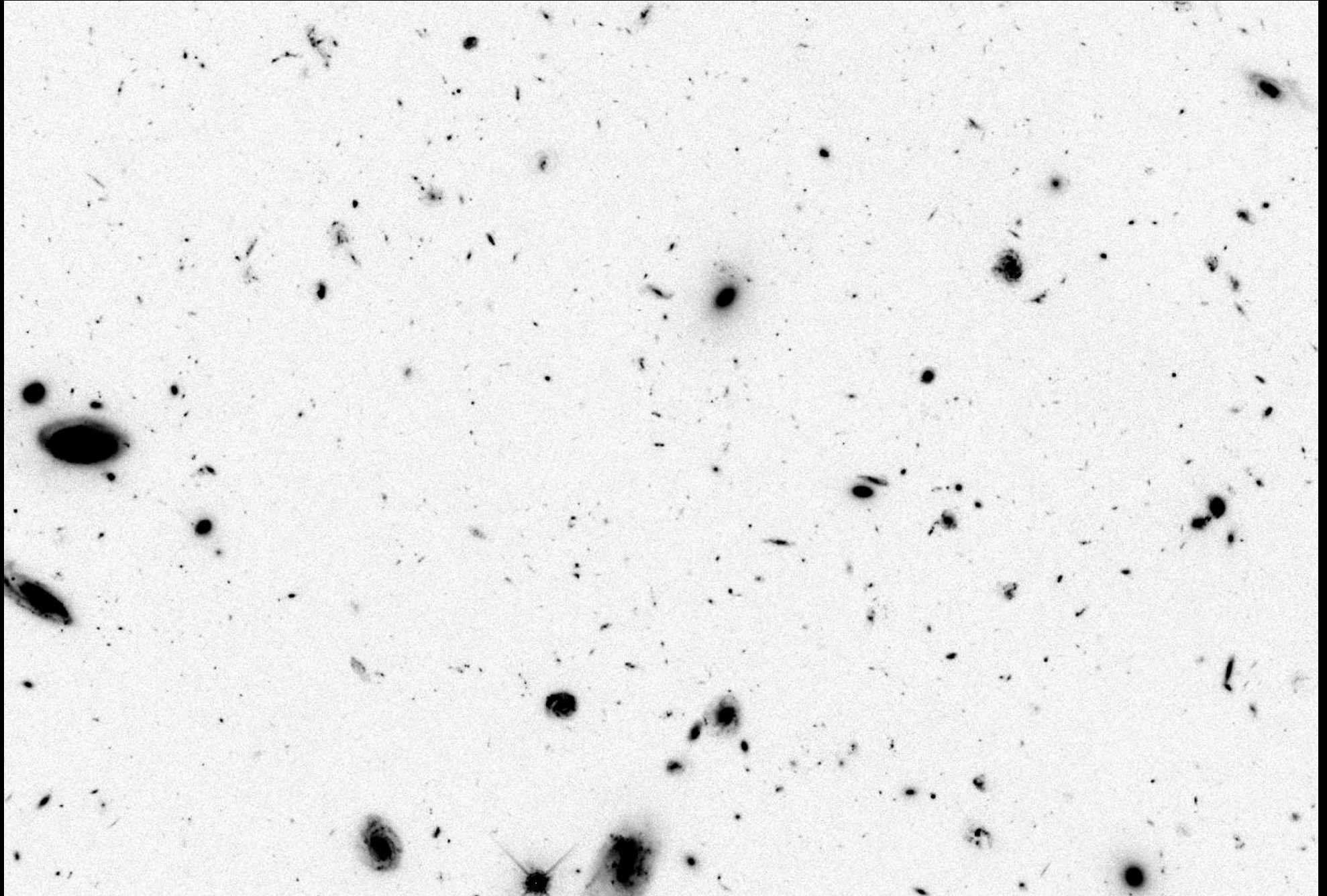
Reliability and Completeness of Object Counts over a factor 10^8 in flux



Data: Ground-based + Hubble object counts from $\lambda \simeq 0.2\text{--}2.0\mu\text{m}$ (Windhorst et al. 2011, *Astrophys. J. Suppl.* 193, 27): Filters: F225W, F275W, F336W, F435W, F606W, F775W, F850LP, F098M/F105W, F125W, F160W.

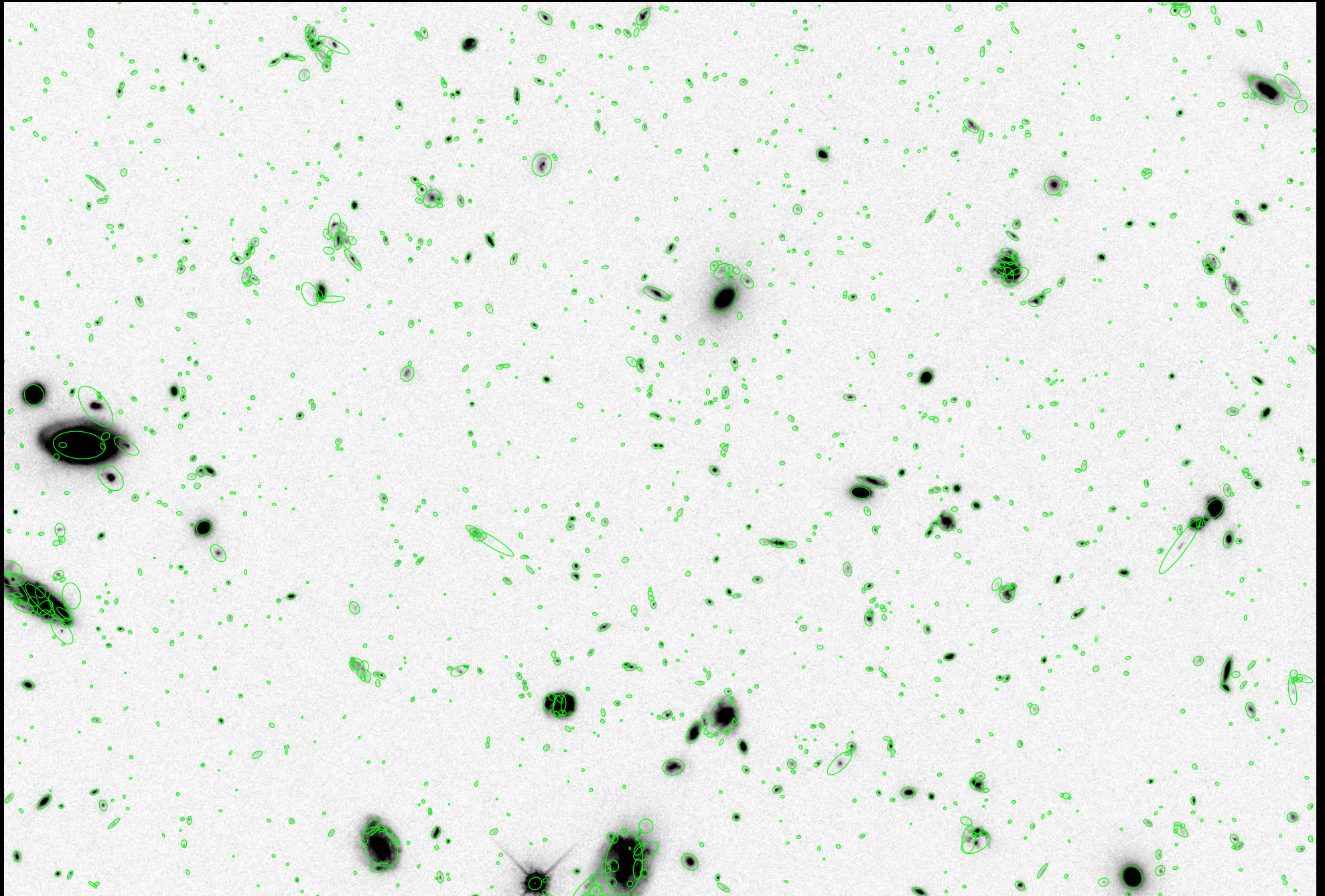
- Reliability and Completeness of Object Counts $\gtrsim 90\%$ compared to the best available cosmological models.

(2a) Object Finding, Detection, Deblending, and Classification method



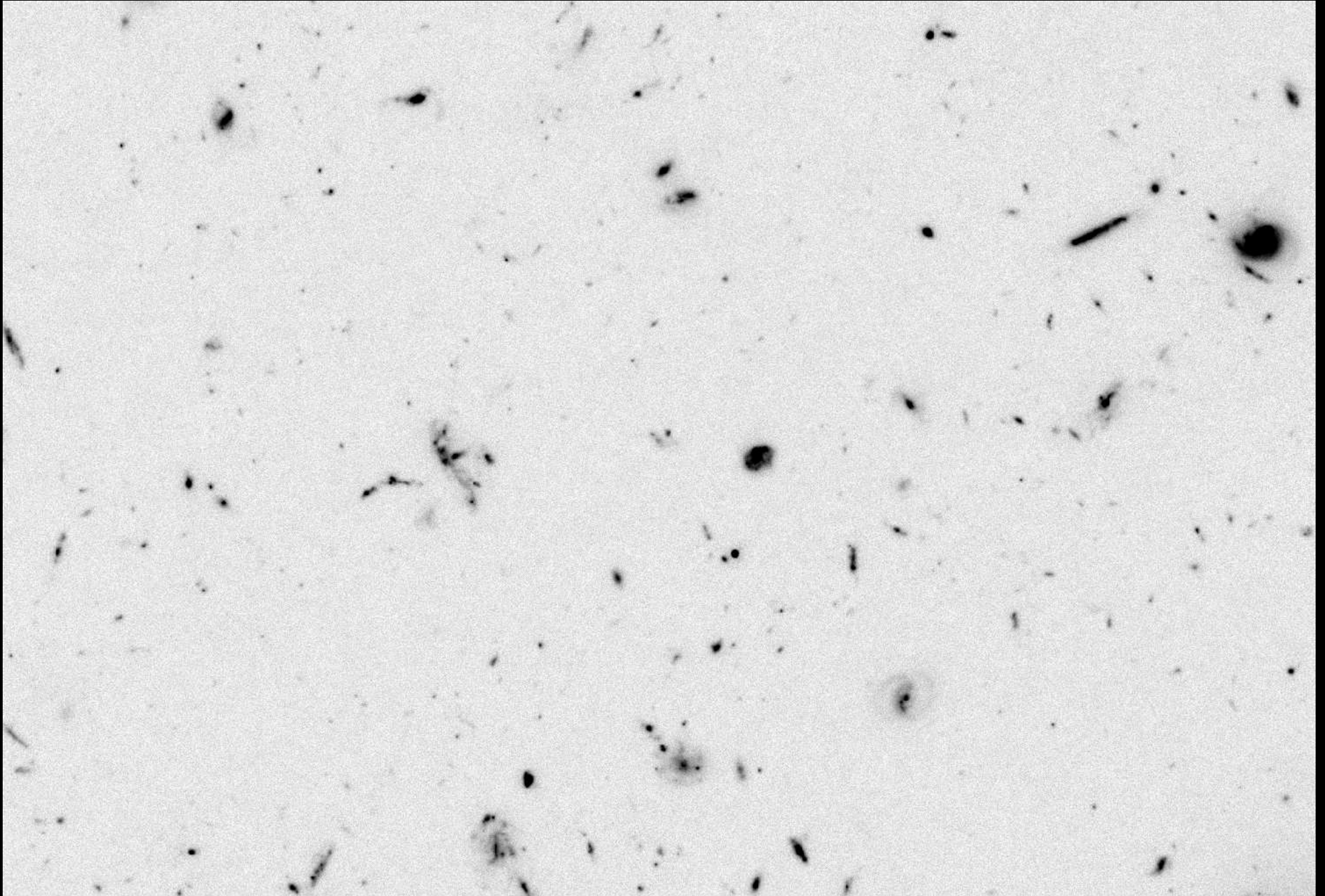
The deepest Hubble images start showing significant galaxy overlap.

(2a) Object Finding, Detection, Deblending, and Classification method



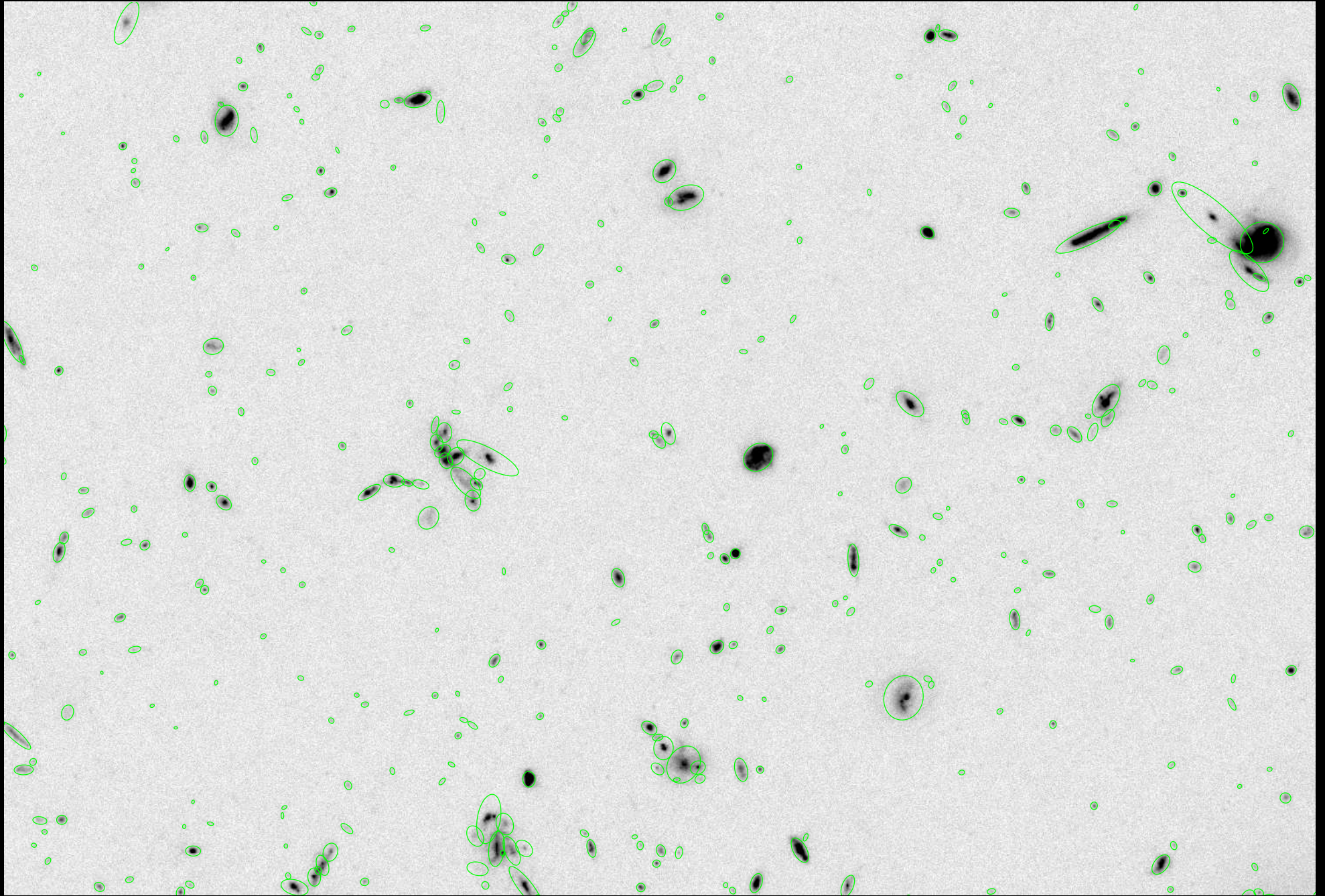
to address crowding, the source-extractor code iteratively deblends the images.

(2a) Object Finding, Detection, Deblending, and Classification method



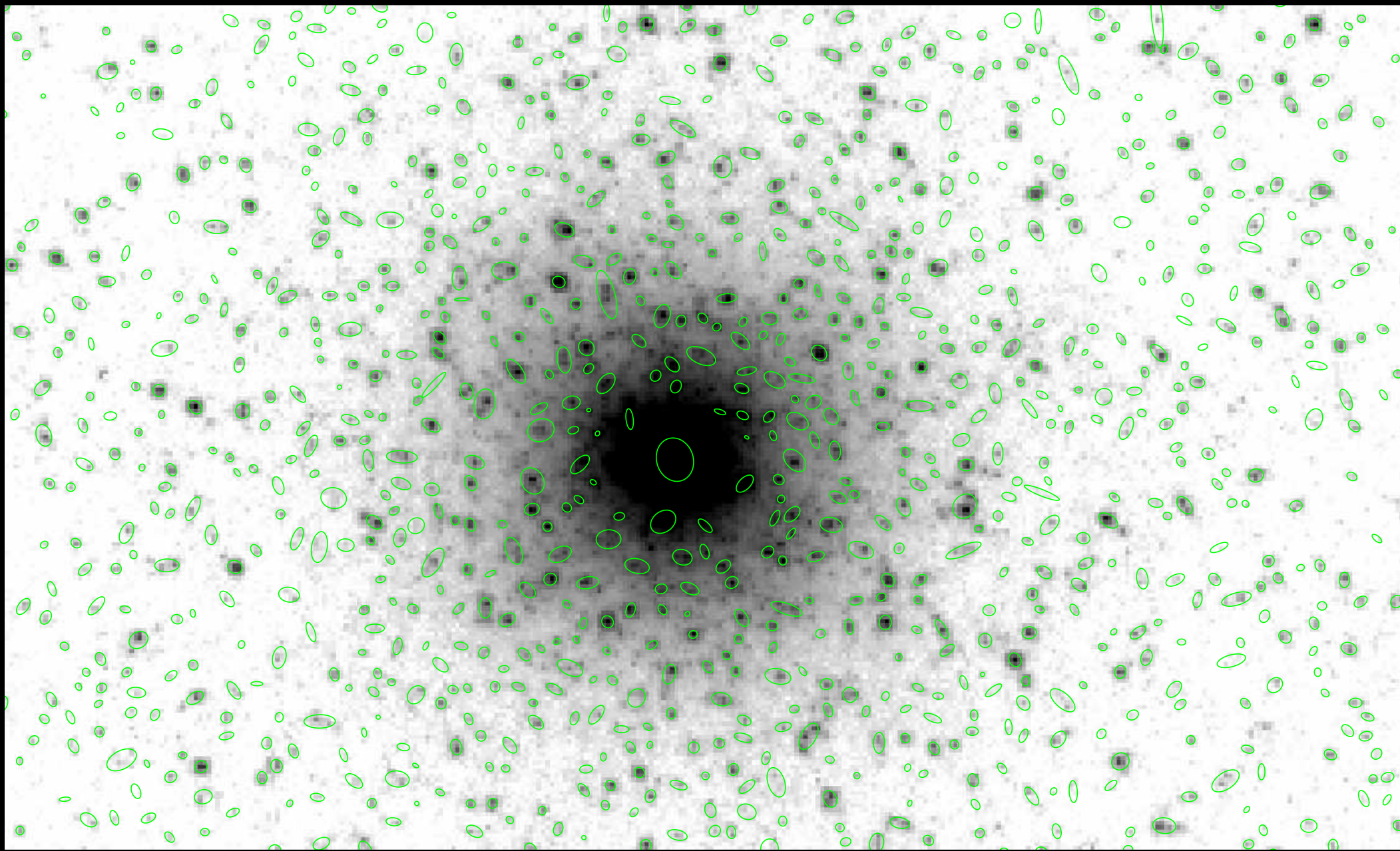
Very complex filamentary galaxies are seen in early stages of formation.

(2a) The Object Finding, Detection, Deblending, and Classification method



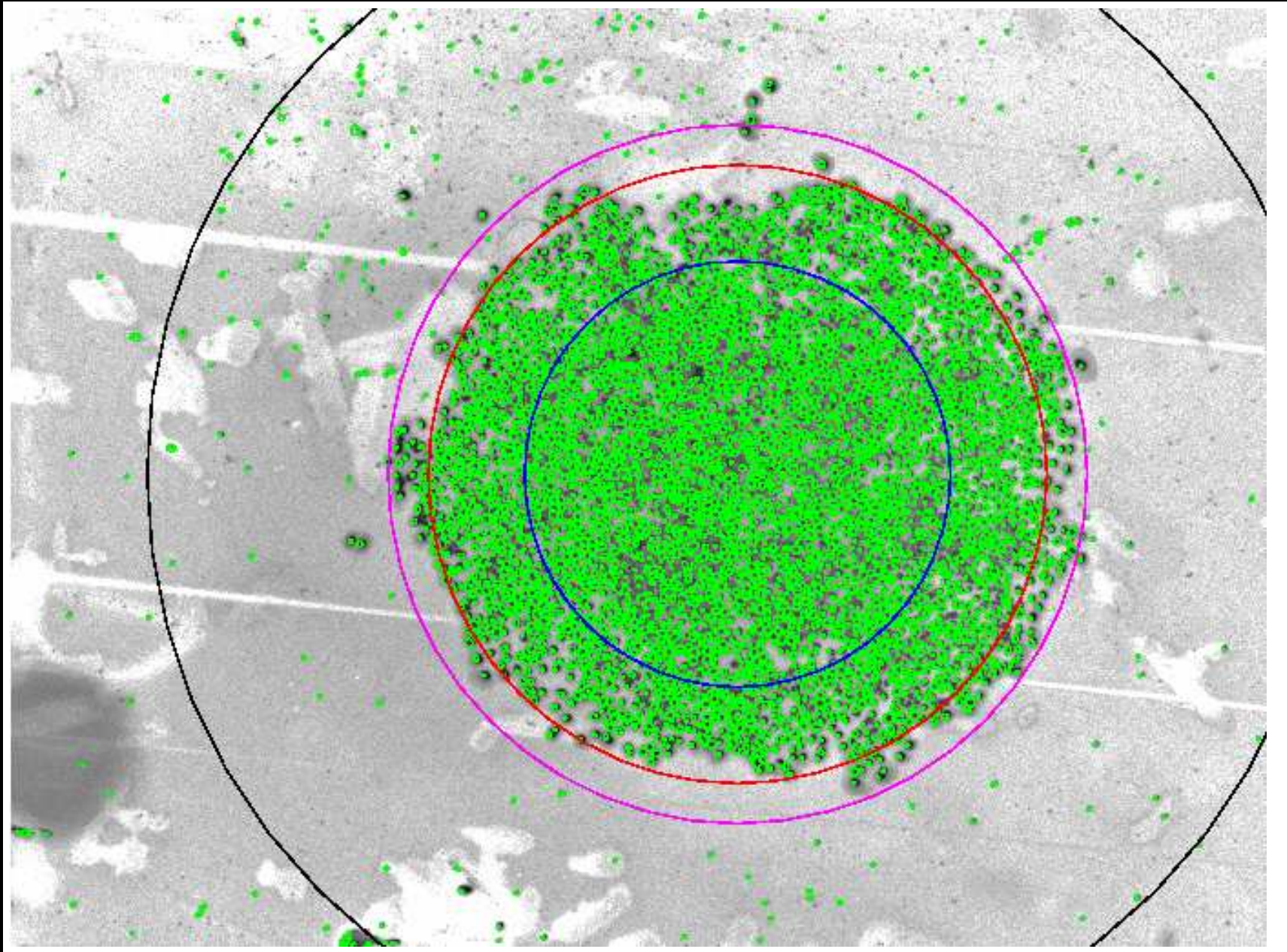
The Source-Extractor software deblends these in chains of elliptical isophotes.

(2b) The Object Finding, Detection, Deblending, and Classification method



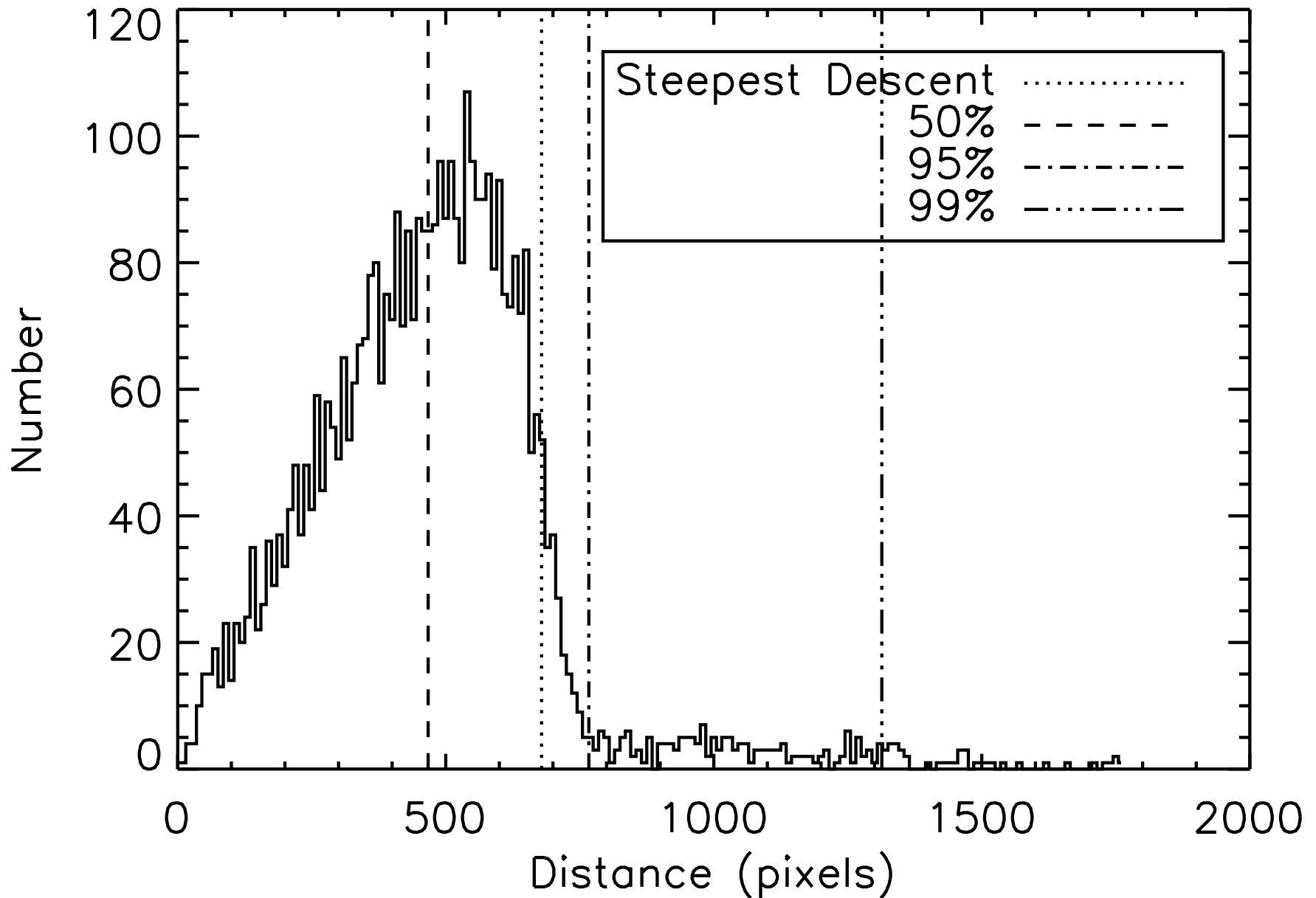
The Source-Extractor algorithm deblends a rich globular cluster of 10^5 stars.
(Only several 10^3 brightest stars plotted, but object overlap is allowed).

(2b) Application of Hubble Software to Study Spreading of Cancer Cells



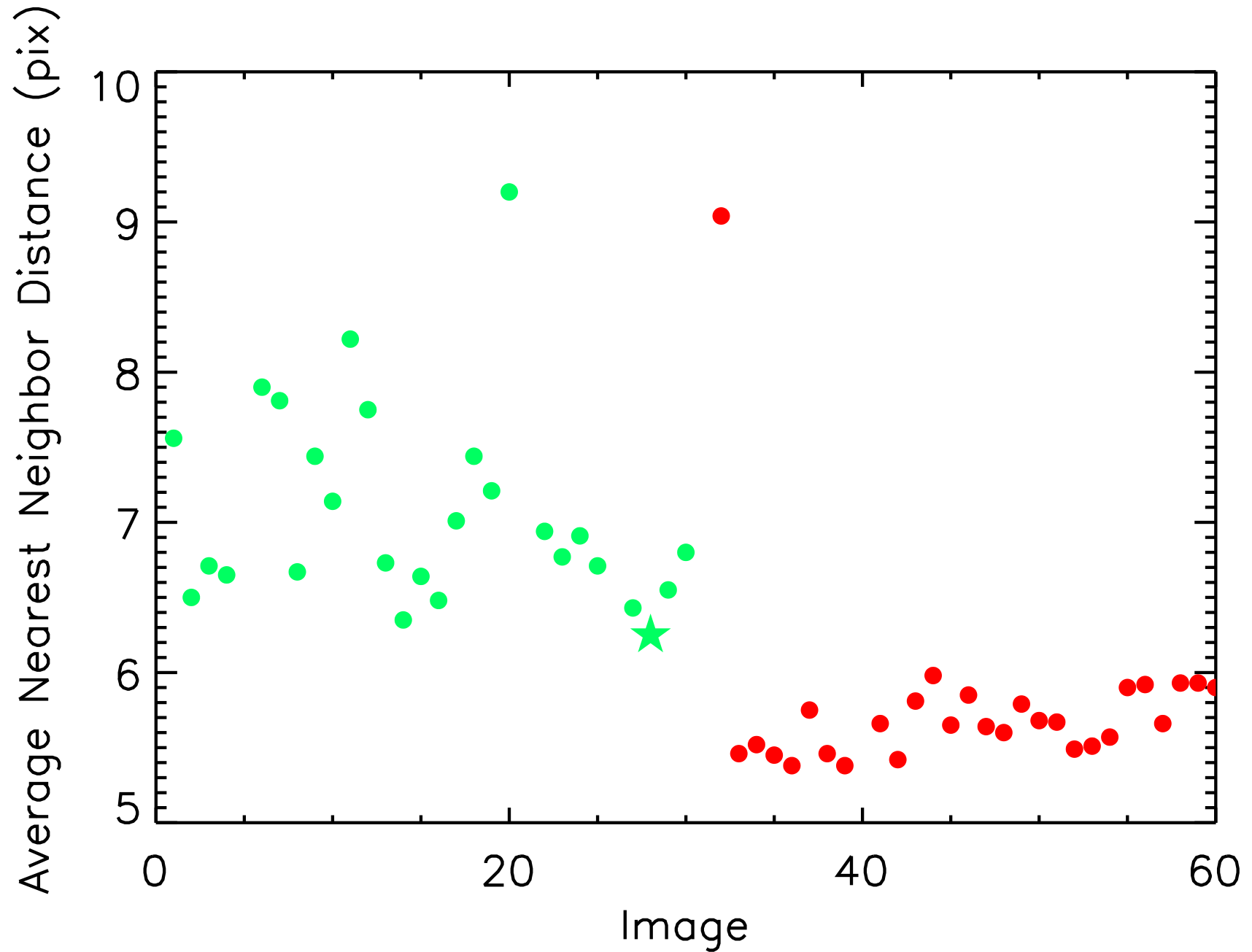
Source-Extractor directly counts number of cells in a cancer biopsy image.
(Collaboration with Dr. M. Berens et al. at TGEN, R. Ryan at ASU.)

(2b) Application of Hubble Software to Study Spreading of Cancer Cells



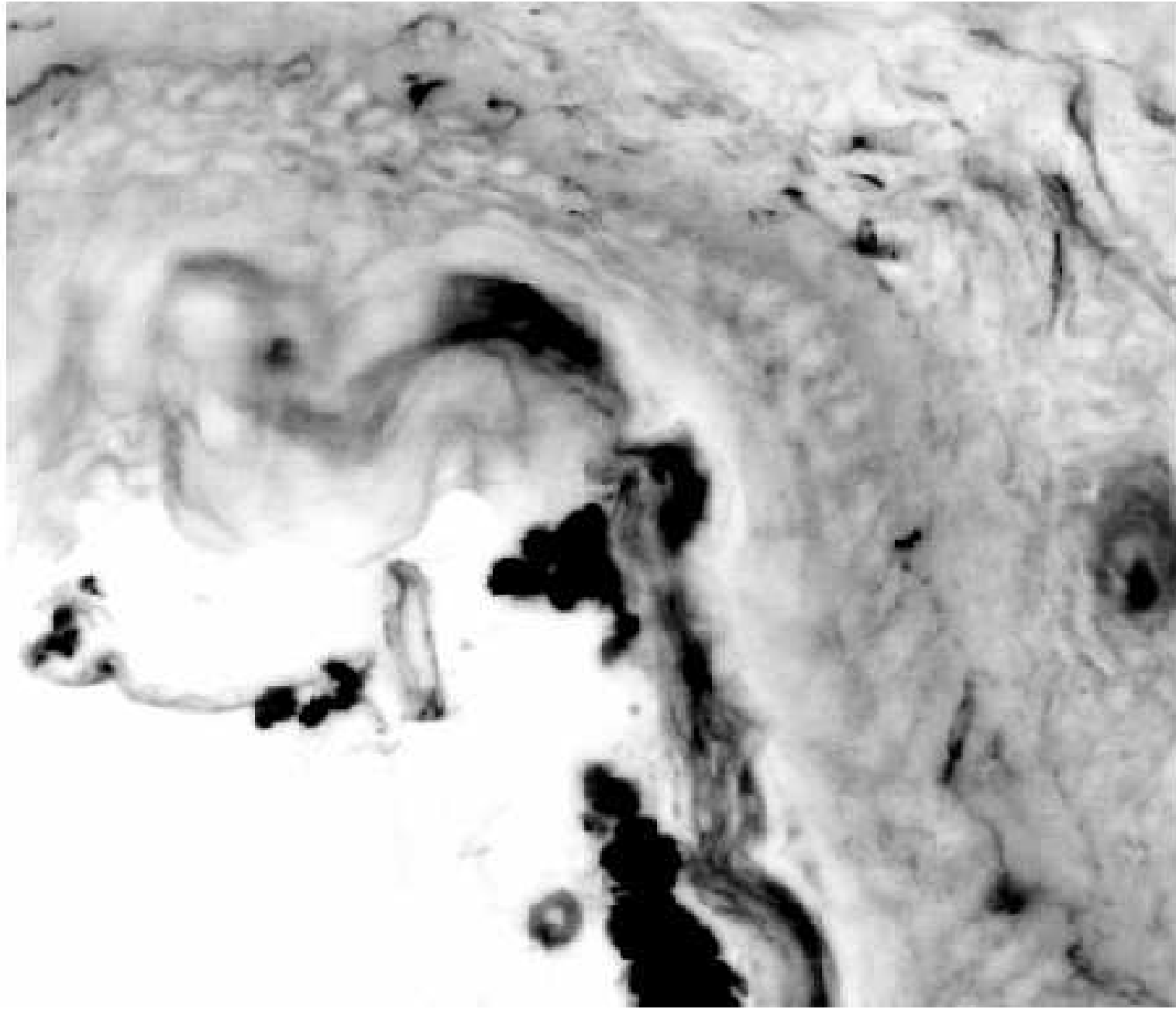
The distribution and spreading of cancer cells is measured quantitatively.

(2b) Application of Hubble Software to Study Spreading of Cancer Cells



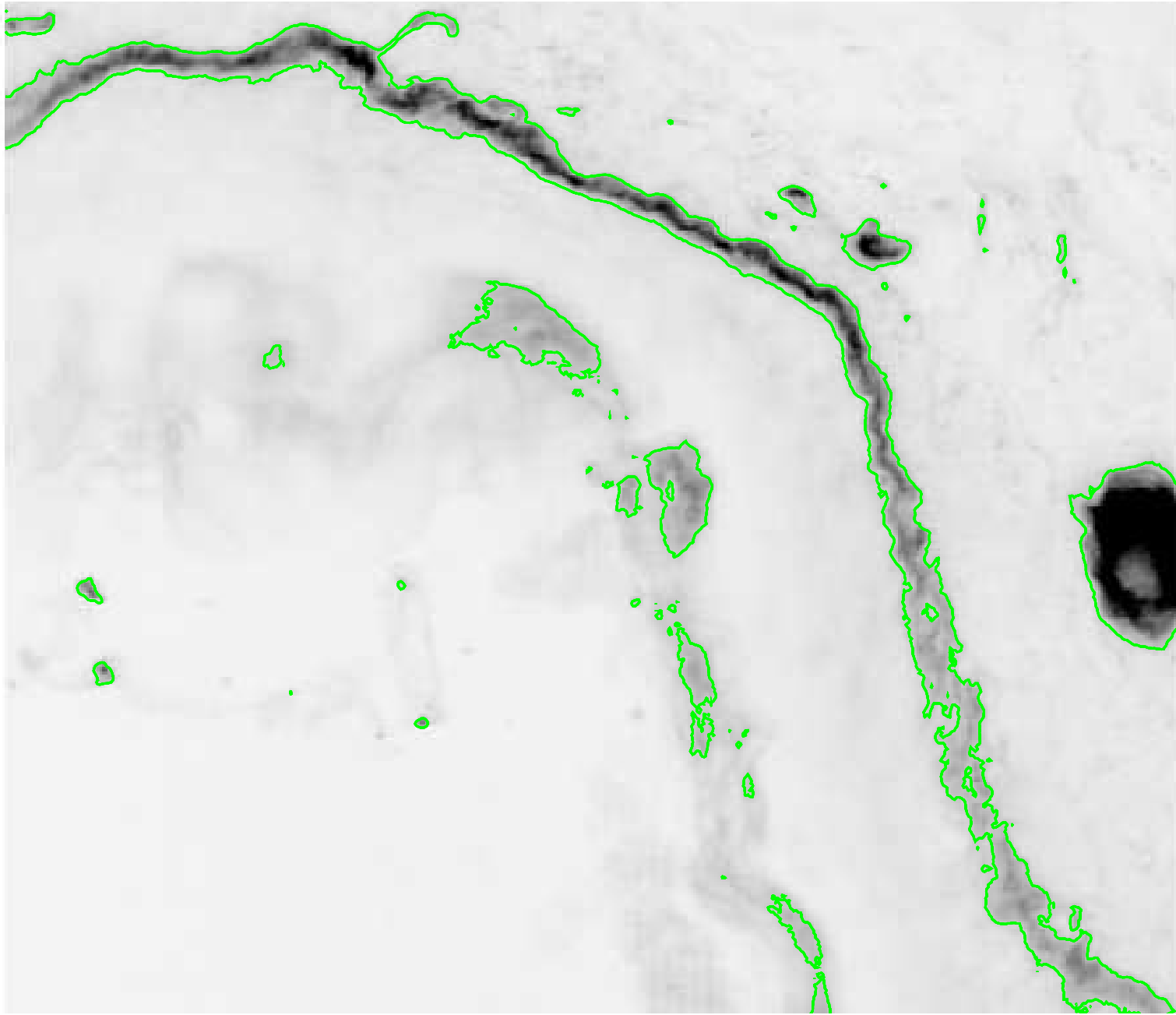
Nearest neighbor analysis bifurcates the spread and non-spread cancer biopsies.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



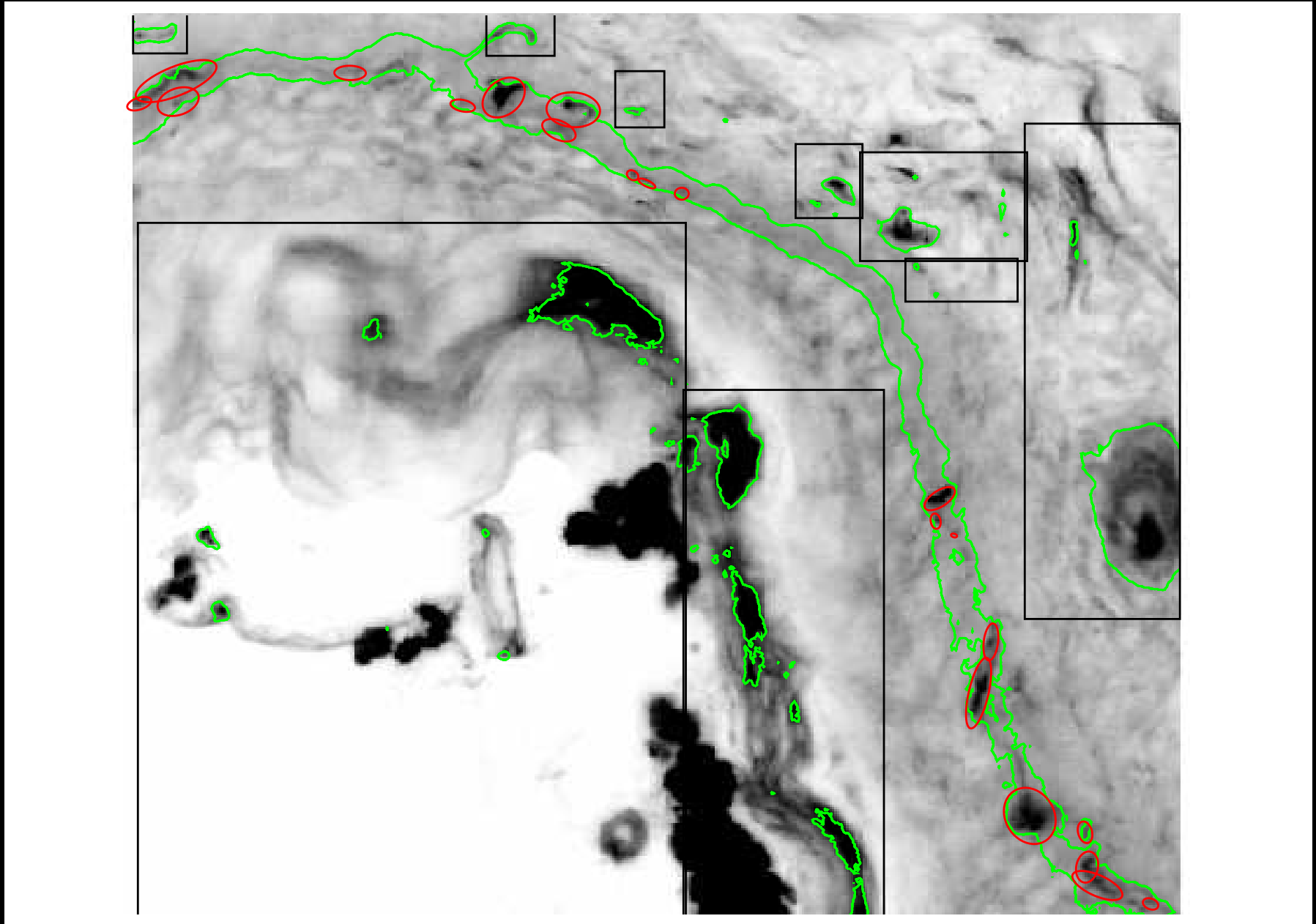
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



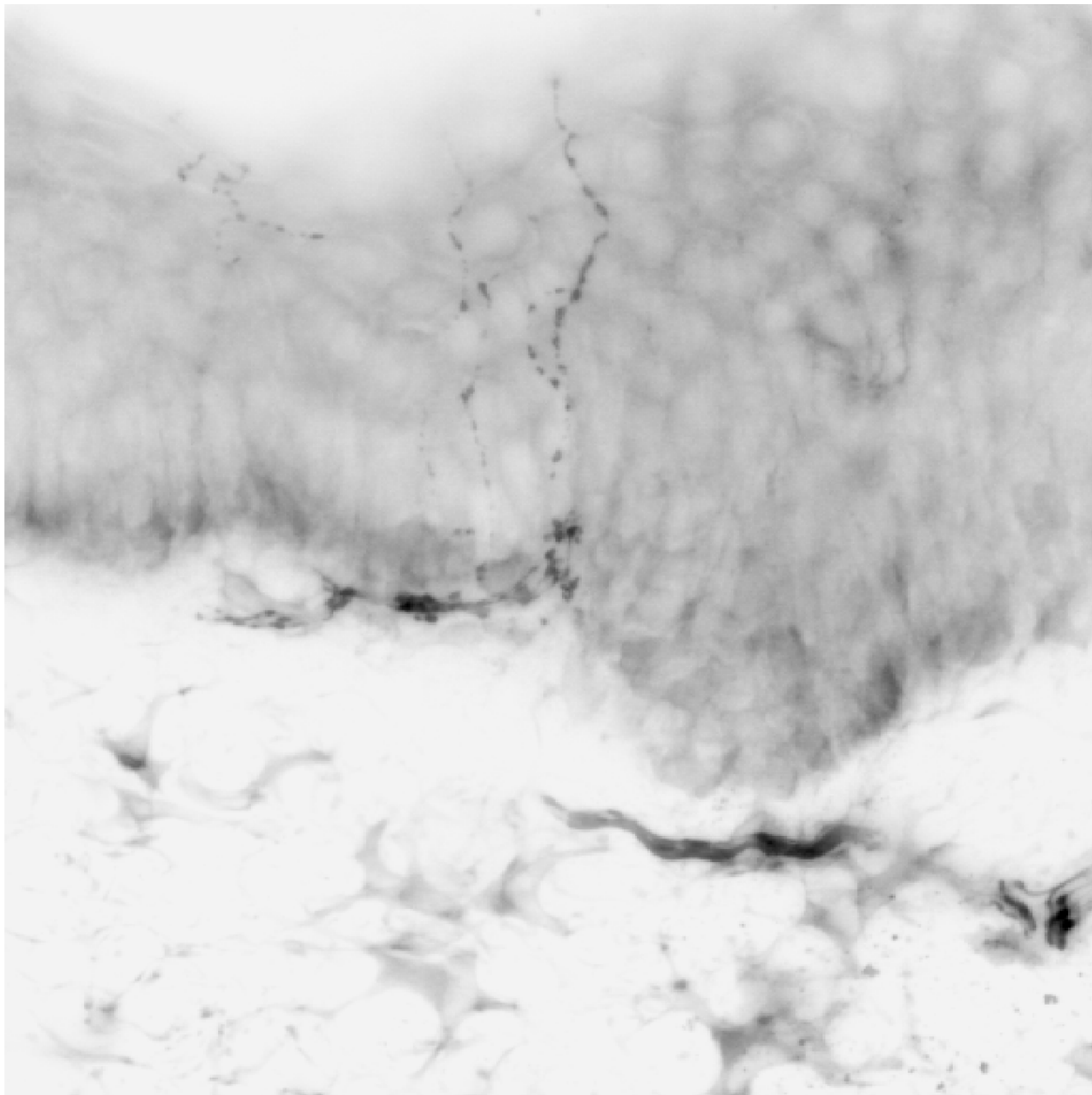
Different staining allows us to define the membrane and other parts of skin.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



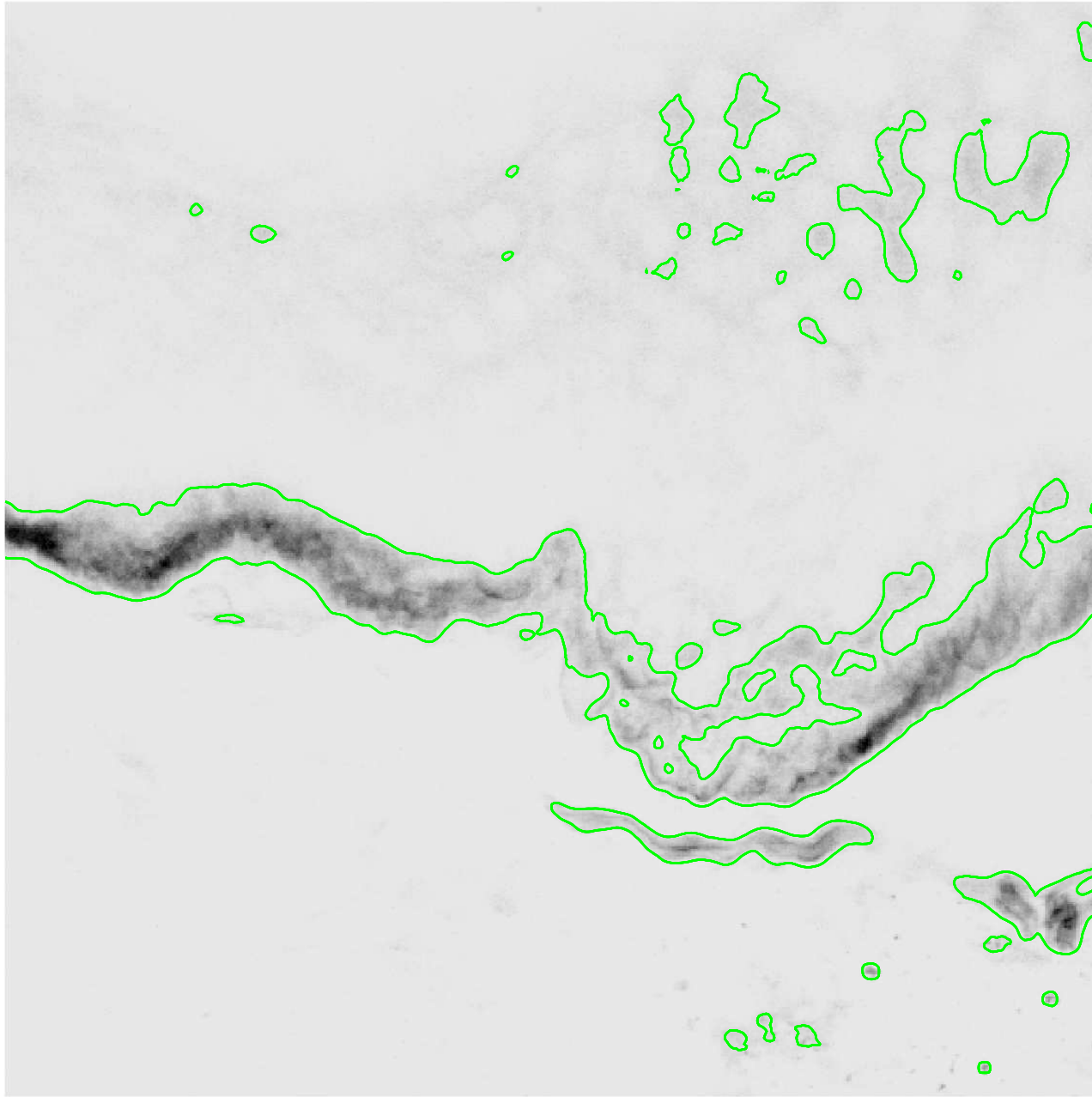
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



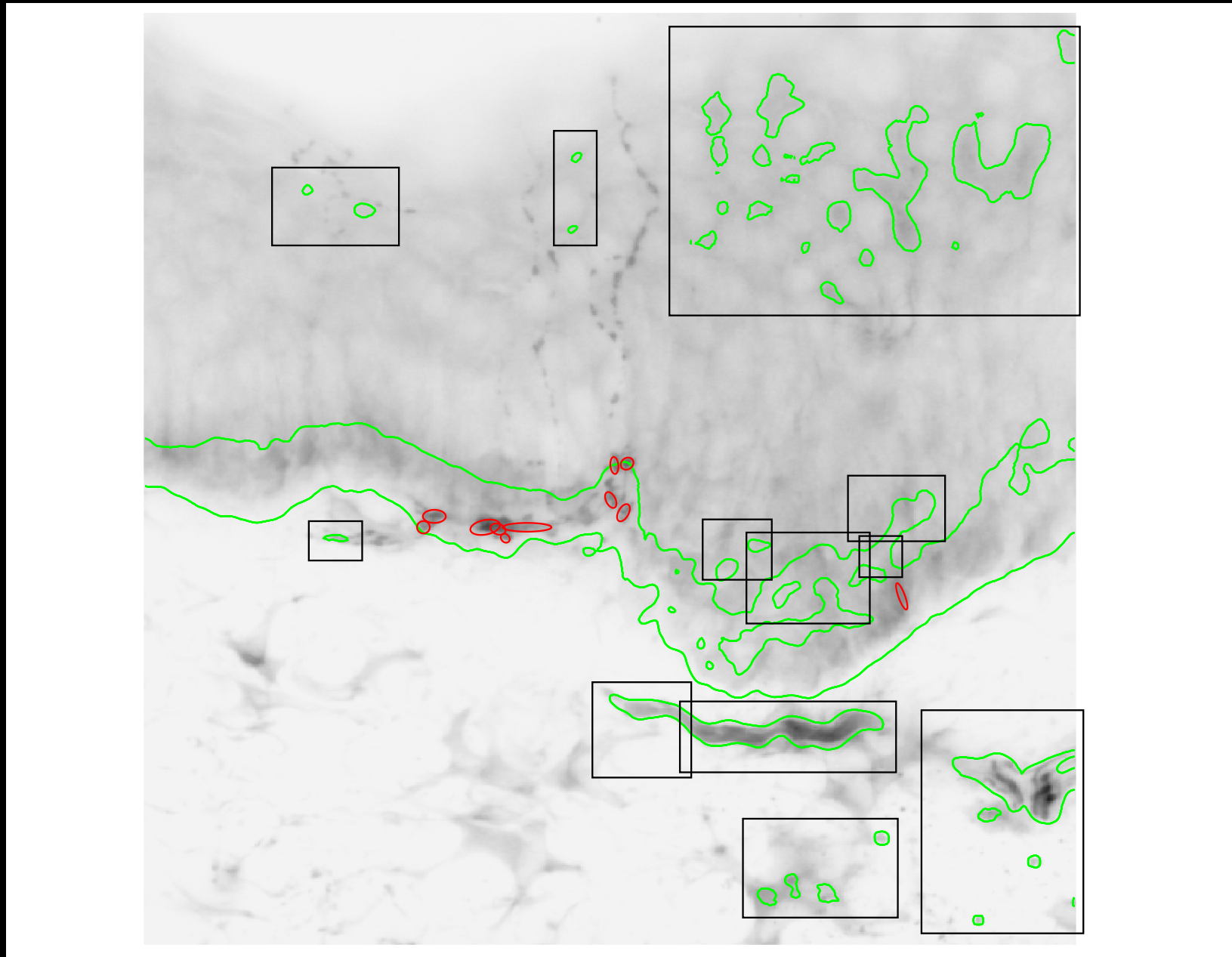
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



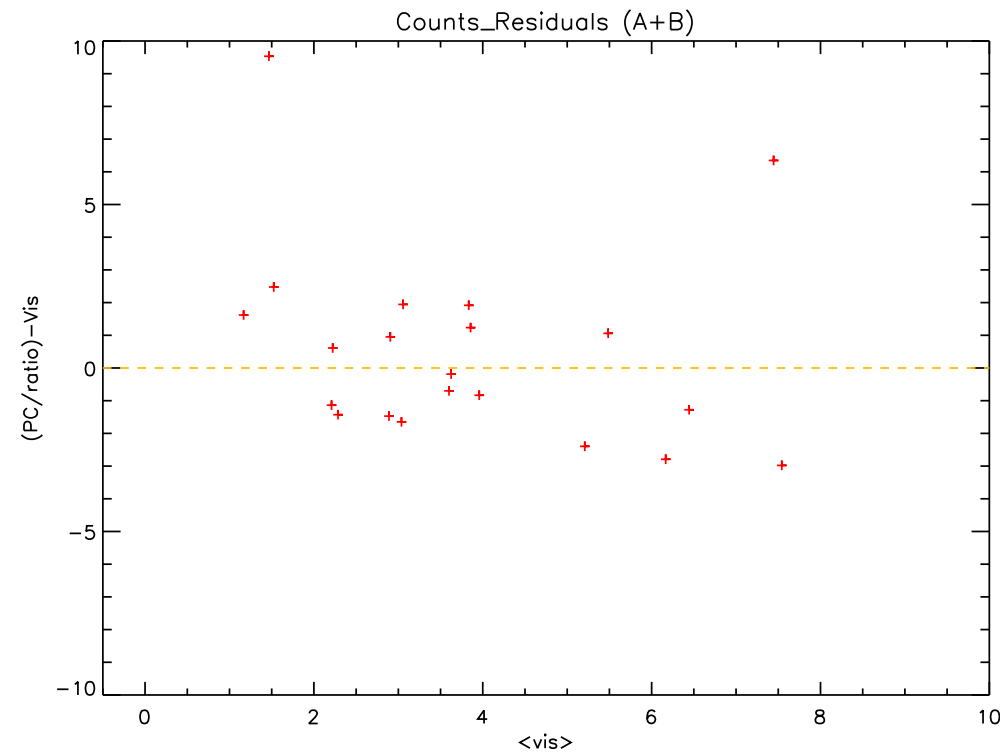
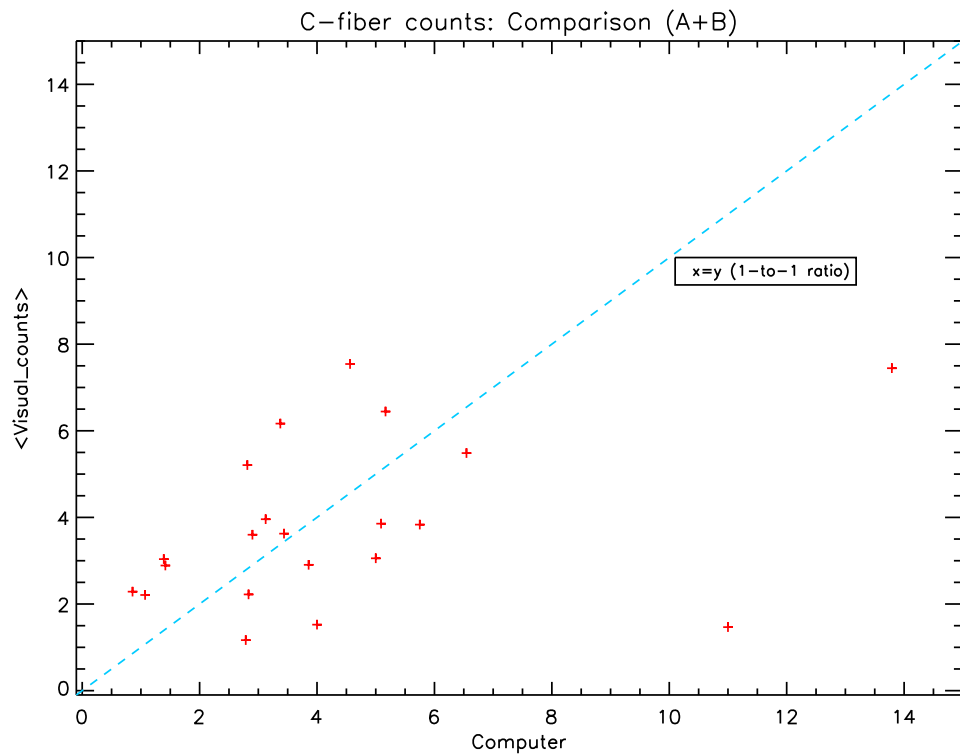
Different staining allows us to define the membrane and other parts of skin.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese Patients



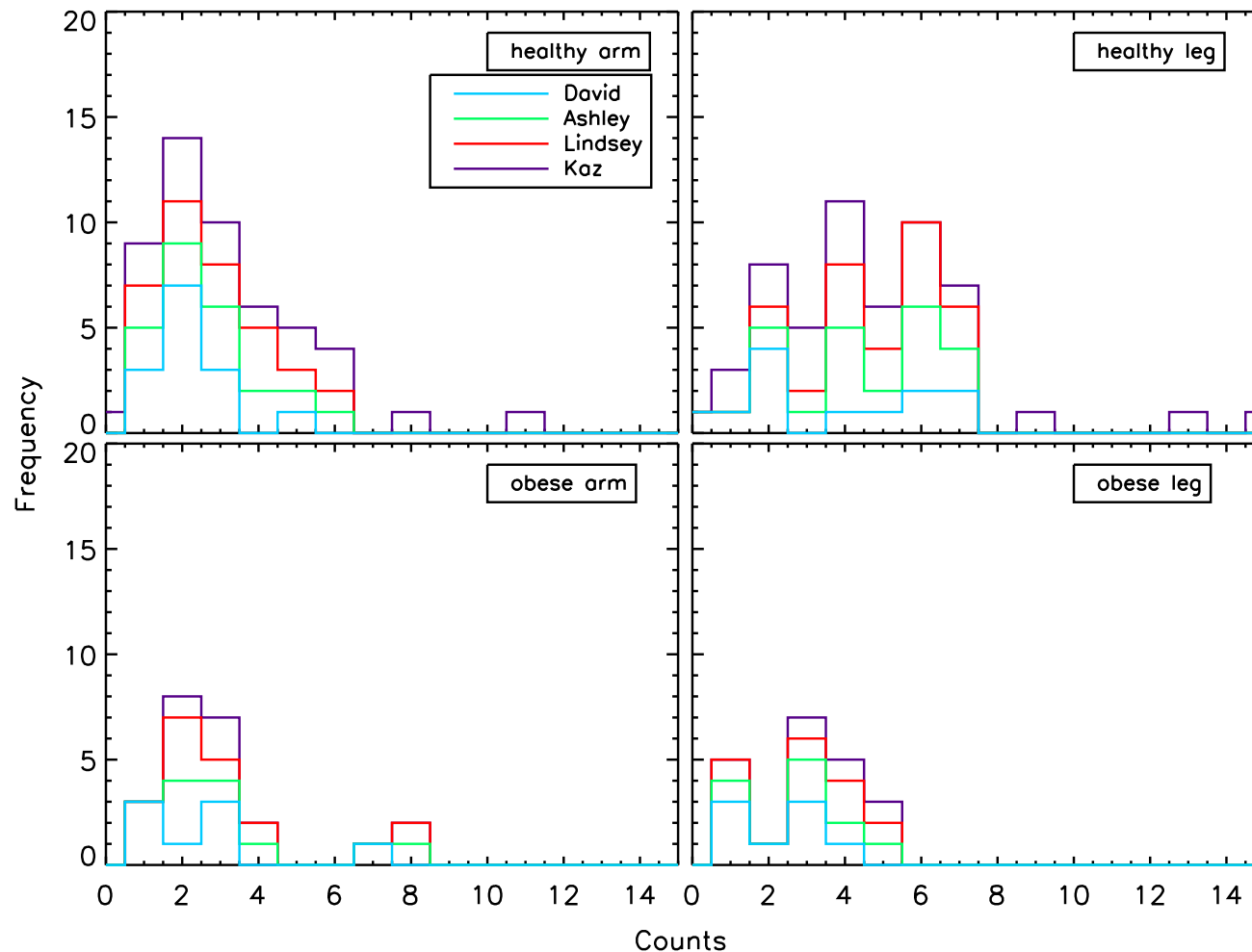
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2c) Application to C-fiber Skin-biopsies



- Machine count correlates well with visual counts, with a few catastrophic failures where software breaks a C-fiber up in too many pieces.
- Average fragmentation is calibrated from expert visual classifiers.
- Catastrophic failures are corrected by hand — working on training the software to recognize and correct those as well.
- Dispersion between machine and visual classifiers is $\sigma \simeq \pm 1.5$ count.

(2c) Application to C-fiber Skin-biopsies



- Cumulative counts of both the machine and the visual classifiers show different distributions for normal and obese patients.
- Obese patients have in general lower C-fiber counts, although the differences between the two populations are not always huge.

(3) Results and Conclusions

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) NASA Hubble Space Telescope image analysis software can detect, deblend and classify overlapping faint galaxy or faint star images.
- (1b) Software is very general — doesn't care what it counts or classifies!

(2) Can use software to identify spreading of Cancer cells or potential markers for Diabetes Type 2 in early stage:

(2a) Galaxy Detection, Deblending, and Classification software is fairly robust and general, minimizing human interaction.

(2b) Application of Hubble Software also allows us to quantitatively study the spreading of Cancer Cells in biopsies.

(2c) Application to C-fiber Skin-biopsies in Normal and Obese patients allows us to quantitatively distinguish healthy from sick patients.

- Reliably find & measure faint features in non-uniform, noisy background.
- Reliably process large numbers of images at once (msec of CPU time).

REFERENCES

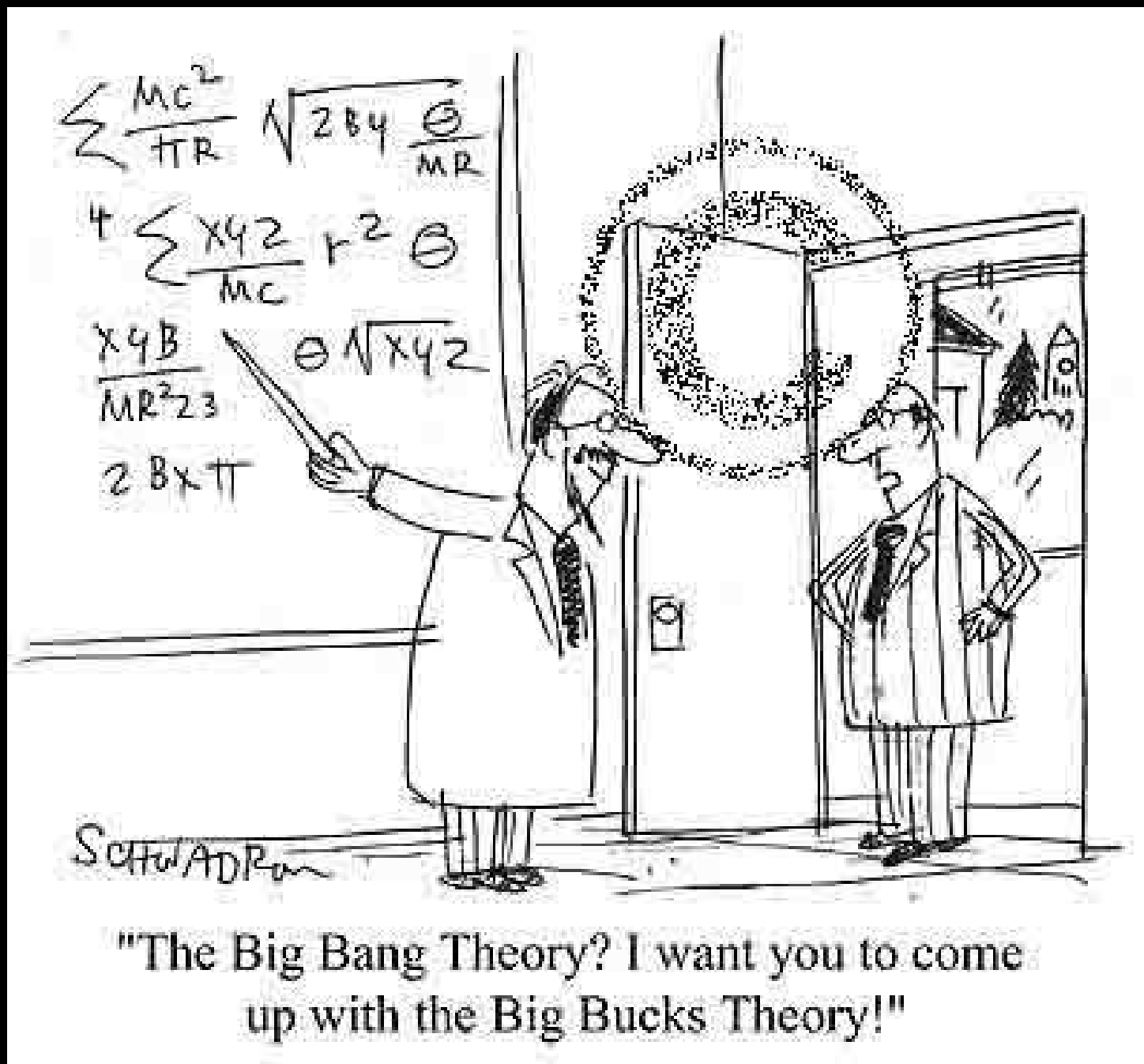
<http://hubblesite.org/newscenter/2010/01/> and [../2011/04/](http://hubblesite.org/newscenter/2011/04/)

Casano, A. R., Brower, J. B., Baluch, D. P., Windhorst, R. A., Targovnik, J. H., & Herman, R. M. 2009, International Journal of Obesity, submitted (Jul. 2011) [<http://www.nature.com/ijo/index.html>] “Small Sensory Nerve Fiber Neuropathy in Morbidly Obese Subjects: Reduced Intra Epidermal Nerve Fiber Density”

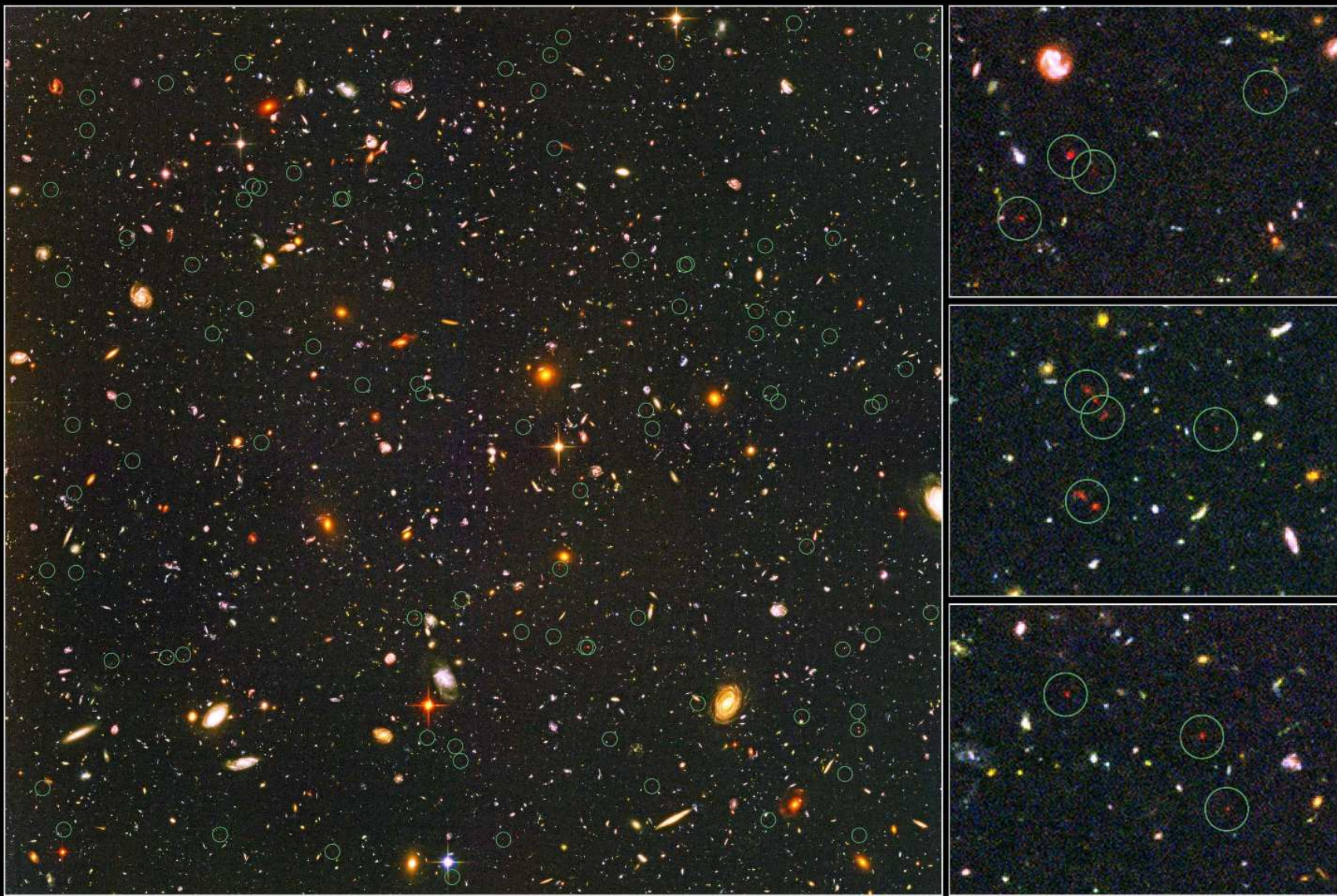
Herman, R. M., Brower, J. B., Stoddard, D. G., Casano, A. R., Targovnik, J. H., Herman, J. H., & Tearse, P. 2006, International Journal of Obesity, 31, 226 [www.nature.com/ijo/journal/v31/n2/full/0803418a.html] “Prevalence of somatic small fiber neuropathy in obesity ”

Tamura, K., Mager, V. A., Burnett, L. A., Olson, J. H., Brower, J. B., Casano, A. R., Baluch, D. P., Targovnik, J. H., Windhorst, R. A., & Herman, R. M. 2009, Journal of Neuroscience Methods, 185, pg. 325–337 [<http://ees.elsevier.com/jneumeth/>] “Semi-Automated Method of Analysis of Small Sensory Nerve Fibers in Human Skin-Biopsies”

SPARE CHARTS



Some cosmologists at ASU decided to step out of the box ...
(although it is hard to please the dean ...)



Distant Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope • Advanced Camera for Surveys

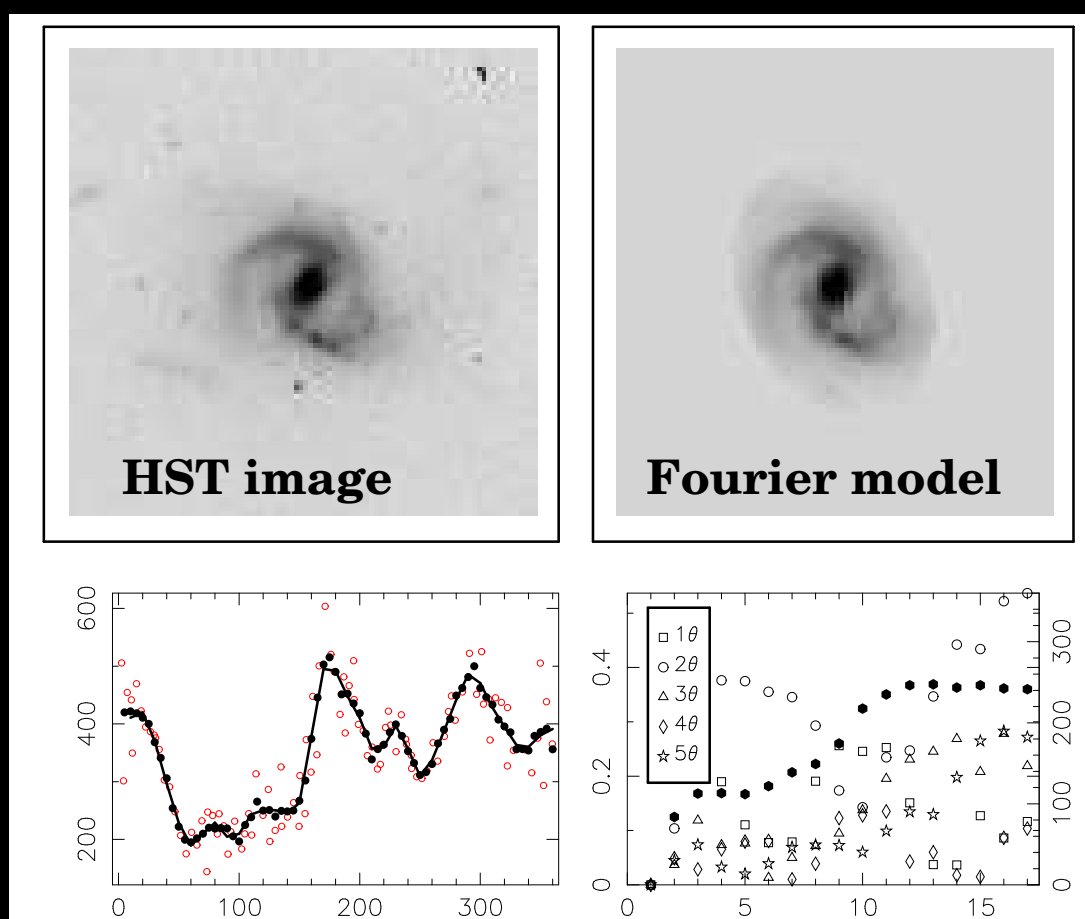
NASA, ESA, R. Windhorst (Arizona State University) and H. Yan (Spitzer Science Center, Caltech)

STScI-PRC04-28

Hubble UltraDeep Field: Dwarf galaxies at $z \simeq 6$ (age $\simeq 1$ Gyr; Yan & Windhorst 2004), many confirmed by spectra at $z \simeq 6$ (Malhotra et al. 2005).

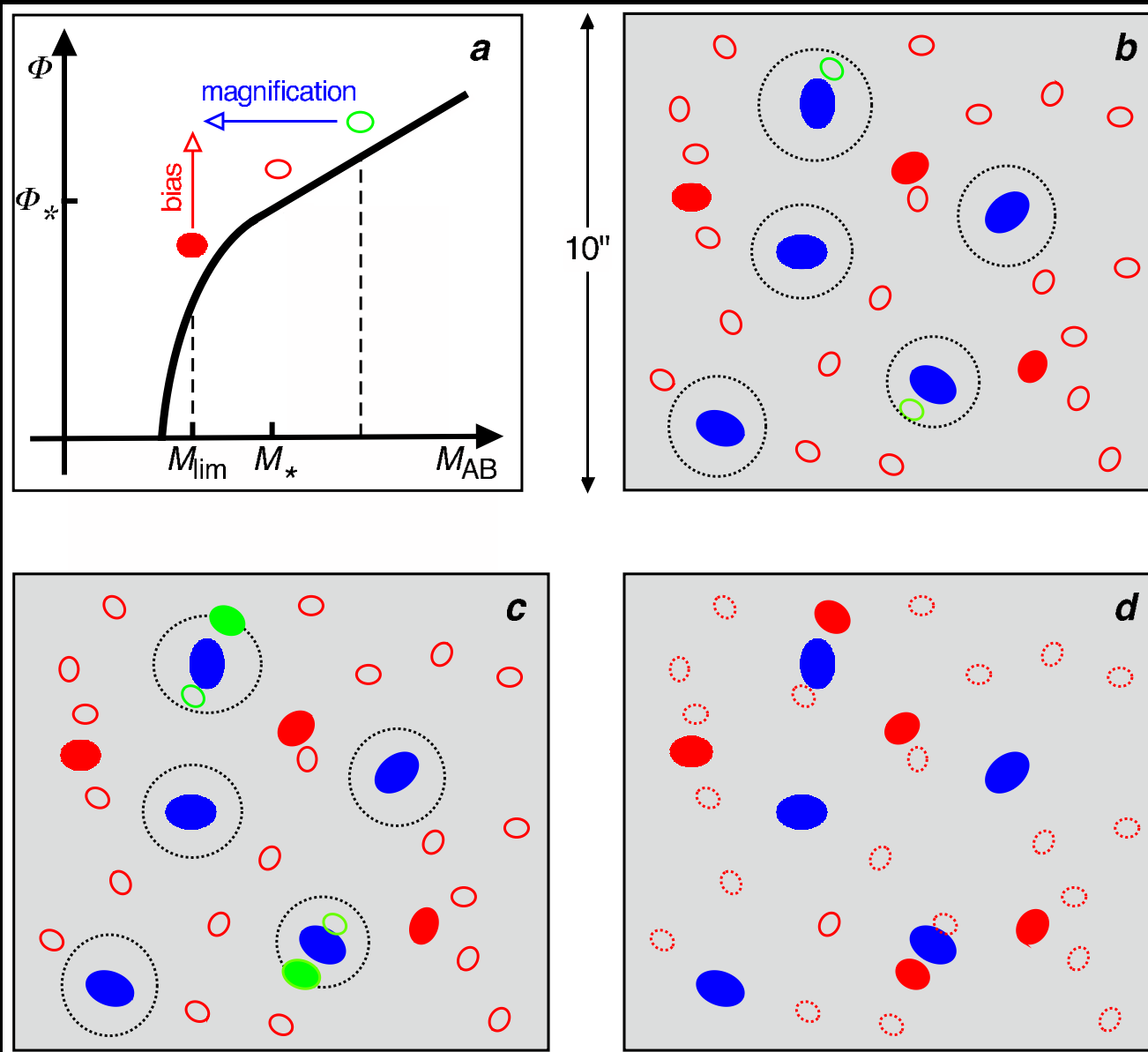


240 hour image of the Hubble UltraDeep Field: 10^4 galaxies $\lesssim 10^{10} \times$
fainter than unaided eye can see, in area smaller than looking through straw!



Fourier Decomposition is a robust way to measure galaxy morphology and structure in a quantitative way (Odewahn et al. 2002):

- (1) Fourier series are made in successive concentric annuli.
- (2) Even Fourier components indicate symmetric parts (arms, rings, bars).
- (3) Odd Fourier components indicate asymmetric parts (lopsidedness).
- (4) JWST can measure the evolution of each feature/class directly.



Hard to see the forest for the trees in the first 0.5 Gyrs?:

- Foreground galaxies ($z \simeq 1-2$ or $\text{age} \simeq 3-6$ Gyr) may gravitationally lens or amplify galaxies at $z \gtrsim 8-10$ ($\text{age} \lesssim 0.5$ Gyr; Wyithe et al. 2011).
- This could change the landscape for JWST observing strategies.